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# Classification Techniques for Predictive Modeling

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# 1. Introduction

In the highly competitive hospitality industry, effective revenue management and operational efficiency are critical for sustaining profitability. One of the major challenges faced by hotels is the uncertainty associated with booking cancellations. When customers cancel their reservations, especially at the last moment, it leads to unoccupied rooms, loss of expected revenue, and inefficient utilization of available resources. This unpredictability not only affects financial performance but also complicates planning related to staffing, inventory management, and service delivery.

When the increasing availability of historical booking data, hotels now have the opportunity to leverage data-driven approaches to better understand customer behavior and predict future outcomes. In this context, predictive analytics plays a vital role in identifying patterns and trends that influence booking cancellations. By accurately predicting whether a booking is likely to be cancelled, hotels can proactively implement strategies such as overbooking, dynamic pricing, targeted promotions, and stricter cancellation policies to mitigate potential losses.

The primary objective of this project is to build and evaluate predictive models that can classify hotel bookings as either cancelled or not cancelled. The dataset used in this analysis contains a wide range of features, including customer-relates attributes (such as number of adults and children), booking characteristics (such as lead time, number of nights, and special requests), pricing information, and market segment details. These variables provide valuable insights into customer preferences and booking behavior.

The analysis begins with data preprocessing and exploratory data analysis to identify key trends, patterns and relationships within the dataset. This includes examining the distribution of bookings, understanding the role of different market segments, and identifying factors that may contribute to higher cancellation rates. Following this, two classification techniques, Logistic Regression and Decision Tree Classifier are applied to model the probability of booking cancellations. These models are evaluated using multiple performance metrics, including accuracy, precision, recall, and F1-score, to ensure a comprehensive assessment of their effectiveness. Special attention is given to the model's ability to correctly identify cancelled bookings, as this is of primary importance from a business perspective.

The insights derived from this study are intended to support hotel management in making informed, data-driven decisions. By understanding the key drivers of booking cancellations and leveraging predictive models, hotels can improve their demand forecasting, optimize pricing strategies, enhance customer engagement, and reduce the financial impact of cancellations. Ultimately, this project demonstrates how data analytics and machine learning techniques can be effectively applied to solve real-world business problems in the hospitality sector.

# 2. Business Problem

## 2.1 Problem Context

In recent years, the hospitality industry has witnessed significant growth, leading to increased competition among hotels. To remain competitive and profitable, hotels must efficiently manage their room inventory, pricing strategies, and customer relationships. One of the major operational challenges faced by hotels is the high rate of booking cancellations. Cancellations can occur due to various reasons such as changes in travel plans, pricing sensitivity, or lack of commitment from customers. These cancellations result in revenue loss, underutilization of available rooms, and inefficiencies in operational planning. With the availability of large volumes of booking data, hotels can leverage data analytics and machine learning techniques to better understand customer behavior and anticipate cancellations in advance. By identifying patterns and trends associated with cancellations, hotels can implement proactive strategies such as overbooking, targeted marketing, and dynamic pricing to minimize financial losses and improve overall efficiency.

## 2.2 Problem Definition

The primary problem addressed in this project in the inability of hotels to accurately predict booking cancellations in advance. Due to this uncertainty, hotels often face challenges in managing room occupancy, forecasting demand, and optimizing revenue. Traditional approaches rely heavily on historical averages or manual judgement, which may not capture the complex relationships between different booking attributes and cancellation behavior. As a result, hotels are unable to effectively identify high-risk bookings, leading to missed opportunities for intervention. Therefore, there is a need for a systematic, data-driven approach that can analyze historical booking data and accurately classify whether a booking is likely to be cancelled or not.

## 2.3 Objective

The main objective of this project is to develop a predictive model that can classify hotel bookings as cancelled or not cancelled based on various features in the dataset.

The specific objectives include:

- To perform data preprocessing and exploratory data analysis to understand the dataset and identify key patterns influencing booking cancellations.
- To build and implement classification models, specifically Logistic Regression and Decision Tree Classifier, for predicting booking status.
- To evaluate the performance of these models using appropriate metrics such as accuracy, precision, recall, and F1-score.
- To identify the most significant factors contributing to booking cancellations.
- To provide actionable business insights and recommendations that can help hotels reduce cancellations and improve operational efficiency.

# 3. Data Description

The dataset used in this project contains information related to hotel bookings, with the objective of predicting whether a booking will be cancelled or not. It consists of a large number of records, where each row represents an individual booking and each column represents a specific attribute associated with that booking. Each variable is described below:

- Booking_ID: Represents the unique identifier of each booking
- no_of_adults: Represents number of adults
- no_of_children: Represents number of children
- no_of_weekend_nights: Represents number of weekend nights (Saturday or Sunday) during which the guest stayed or booked to stay at the hotel.
- no_of_week_nights: Represent the number of week nights (Monday to Friday) during which the guest stayed or booked to stay at the hotel.
- Type_of_meal_plan: Represents the type of meal plan booked by the customer:
    - Not Selected: Indicates no meal plan selected
    - Meal Plan 1: Indicates Breakfast
    - Mean Plan 2: Indicates half board (breakfast and one other meal)
    - Meal Plan 3: Indicates full board (breakfast, lunch, and dinner)
- required_car_parking_space: Represents whether the customer require a car parking space (0: No, 1: Yes).
- room_type_reserved: Indicates the type of room reserved by the customer. The values are ciphered by INN Hotel Group.
- lead_time: Indiactes the number of days between the date of booking and the arrival date.
- arrival_year: Represents the year of arrival date.
- arrival_month: Represents the month of arrival date.
- arrival_date: Represents the date of the month.
- market_segment_type: Represents the market segment designation.
- repeated_guest: Represents whether the customer is a repeated guest (0: No, 1: Yes)
- no_of_previous_cancellations: Represents the number of previous bookings that were canceled by the customer prior to the current booking.
- no_of_previous_bookings_not_canceled: Represents the number of previous bookings that are not canceled by the customer prior to the current booking.
- avg_price_per_room: Represents the average price of the reservation per day, prices of the rooms are dynamic (in euros).
- no_of_special_requests: Represents the total number of special requests made by the customer.
- booking_status: This is the target variable that indicates whether the booking was canceled or not.

# 4. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the structure of the dataset, identify patterns, and gain initial insights into the variables affecting booking cancellations. This step also helped in preparing the data for modeling and uncovering key relationships between features.

## 4.1 Checking the Structure of Data

### 4.1.1 Initial Data Preview

A preliminary examination of the dataset provides an overview of the structure and variables involved. Each record represents a hotel booking instance, containing information related to customer details, booking preferences, and reservation characteristics. This initial inspection helps in understanding the type and format of the data before further analysis.

The first five rows of the dataset are displayed below to observe the structure and sample values:

| Booking | no_of_adu | no_of_chil | no_of_weekend_ | no_of_week_n | type_of_meal_pl | required_car_parking | room_type_rese | lead_ti | arrival_y | arrival_mo | arrival_d | market_segment | repeated_gu | no_of_previous_cance | no_of_previous_bookings_not | avg_price_per | no_of_special_re | booking_st |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| INN00001 | 2 | 0 | 1 | 2 | Meal Plan 1 | 0 | Room_Type 1 | 224 | 2017 | 10 | 2 | Offline | 0 | 0 | 0 | 65 | 0 | Not_Canceled |
| INN00002 | 2 | 0 | 2 | 3 | Not Selected | 0 | Room_Type 1 | 5 | 2018 | 11 | 6 | Online | 0 | 0 | 0 | 106.68 | 1 | Not_Canceled |
| INN00003 | 1 | 0 | 2 | 1 | Meal Plan 1 | 0 | Room_Type 1 | 1 | 2018 | 2 | 28 | Online | 0 | 0 | 0 | 60 | 0 | Canceled |
| INN00004 | 2 | 0 | 0 | 2 | Meal Plan 1 | 0 | Room_Type 1 | 211 | 2018 | 5 | 20 | Online | 0 | 0 | 0 | 100 | 0 | Canceled |
| INN00005 | 2 | 0 | 1 | 1 | Not Selected | 0 | Room_Type 1 | 48 | 2018 | 4 | 11 | Online | 0 | 0 | 0 | 94.5 | 0 | Canceled |

The last five rows of the dataset are displayed below to ensure data consistency and check for any irregularities at the end of the dataset:

| Booking_ | no_of_adul | no_of_childr | no_of_weekend_nig | no_of_week_nig | type_of_meal_pl | required_car_parking_sp | room_type_reserv | lead_tim | arrival_ye | arrival_mont | arrival_dat | market_segment_ty | repeated_gu | no_of_previous_cancellat | no_of_previous_bookings_not_can | avg_price_per_ro | no_of_special_requ | booking_stat |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| INN36271 | 3 | 0 | 2 | 6 | Meal Plan 1 | 0 | Room_Type 4 | 85 | 2018 | 8 | 3 | Online | 0 | 0 | 0 | 167.8 | 1 | Not_Canceled |
| INN36272 | 2 | 0 | 1 | 3 | Meal Plan 1 | 0 | Room_Type 1 | 228 | 2018 | 10 | 17 | Online | 0 | 0 | 0 | 90.95 | 2 | Canceled |
| INN36273 | 2 | 0 | 2 | 6 | Meal Plan 1 | 0 | Room_Type 1 | 148 | 2018 | 7 | 1 | Online | 0 | 0 | 0 | 98.39 | 2 | Not_Canceled |
| INN36274 | 2 | 0 | 0 | 3 | Not Selected | 0 | Room_Type 1 | 63 | 2018 | 4 | 21 | Online | 0 | 0 | 0 | 94.5 | 0 | Canceled |
| INN36275 | 2 | 0 | 1 | 2 | Meal Plan 1 | 0 | Room_Type 1 | 207 | 2018 | 12 | 30 | Offline | 0 | 0 | 0 | 161.67 | 0 | Not_Canceled |

Observation:

- The dataset contains a mix of categorical and numerical variables.
- Each row corresponds to a single booking record.
- The structure appears consistent across the dataset.
- No immediate anomalies are visible in the initial preview.

### 4.1.2 Data Shape

```
(36275, 19)
```

- The dataset contains 36,275 rows and 19 columns.

### 4.1.3 Viewing Column Names

```
Index(['Booking_ID', 'no_of_adults', 'no_of_children', 'no_of_weekend_nights',
       'no_of_week_nights', 'type_of_meal_plan', 'required_car_parking_space',
       'room_type_reserved', 'lead_time', 'arrival_year', 'arrival_month',
       'arrival_date', 'market_segment_type', 'repeated_guest',
       'no_of_previous_cancellations', 'no_of_previous_bookings_not_canceled',
       'avg_price_per_room', 'no_of_special_requests', 'booking_status'],
      dtype='object')
```

- The dataset consists of 19 variables, capturing different aspects of hotel bookings such as customer details, booking behavior, and pricing.
- Features include demographic indicators (e.g., number of adults, children), booking details (e.g., lead time, arrival date, number of nights), and customer history (e.g., previous cancellations, repeated guest).
- It also contains categorical variables like type of meal plan, room type reserved, and market segment type, which provide insights into customer preferences.
- Financial information is represented through variables such as average price per room.
- The column **booking_status** serves as the target variable, indicating whether a booking was canceled or not.
- The presence of a unique identifier, **Booking_ID,** suggests that each record represents a distinct booking instance.

### 4.1.4 Data Background and Content

| # | Column | Non-NULL Count | Dtype |
|---|---|---|---|
| 0 | Booking_ID | 36275 non-NULL | object |
| 1 | no_of_adults | 36275 non-NULL | int64 |
| 2 | no_of_children | 36275 non-NULL | int64 |
| 3 | no_of_weekend_nights | 36275 non-NULL | int64 |
| 4 | no_of_week_nights | 36275 non-NULL | int64 |
| 5 | type_of_meal_plan | 36275 non-NULL | object |
| 6 | required_car_parking_space | 36275 non-NULL | int64 |
| 7 | room_type_reserved | 36275 non-NULL | object |
| 8 | lead_time | 36275 non-NULL | int64 |
| 9 | arrival_year | 36275 non-NULL | int64 |
| 10 | arrival_month | 36275 non-NULL | int64 |
| 11 | arrival_date | 36275 non-NULL | int64 |
| 12 | market_segment_type | 36275 non-NULL | object |
| 13 | repeated_guest | 36275 non-NULL | int64 |
| 14 | no_of_previous_cancellations | 36275 non-NULL | int64 |
| 15 | no_of_previous_bookings_not_canceled | 36275 non-NULL | int64 |
| 16 | avg_price_per_room | 36275 non-NULL | float64 |
| 17 | no_of_special_requests | 36275 non-NULL | int64 |
| 18 | booking_status | 36275 non-NULL | object |

- The dataset contains 36,275 entries and 19 columns, confirming a moderately large dataset suitable for predictive modeling.
- All columns show 36,275 non-null values, indicating that no missing values are present in the dataset.
- The dataset consists of a mix of data types:
  - 13 integer (int64) variables: representing count-based features such as the number of adults, children, nights, previous cancellations, and special requests.
  - 1 float (float64) variable: avg_price_per_room, representing continuous pricing information.
  - 5 categorical (object) variables: including meal plan type, room type, market segment type, and booking status.
- The presence of both categorical and numerical features suggests that data preprocessing (encoding) is required before applying machine learning models.
- The dataset memory usage is relatively low (5.3+ MB), indicating efficient storage and ease of processing.

### 4.1.5 Checking NULL values

```
Booking_ID                            0
no_of_adults                          0
no_of_children                        0
no_of_weekend_nights                  0
no_of_week_nights                     0
type_of_meal_plan                     0
required_car_parking_space            0
room_type_reserved                    0
lead_time                             0
arrival_year                          0
arrival_month                         0
arrival_date                          0
market_segment_type                   0
repeated_guest                        0
no_of_previous_cancellations          0
no_of_previous_bookings_not_canceled  0
avg_price_per_room                    0
no_of_special_requests                0
booking_status                        0
```

- All columns in the dataset contain 0 missing values, indicating that the dataset is complete and clean.
- No imputation or data filling techniques are required, which helps maintain the original integrity of the data.
- Since there are no missing values, the dataset is ready for preprocessing and modeling without further data cleaning.

- This improves model reliability, as missing values can often introduce bias or reduce predictive performance.

## 4.2 Checking Data Types of Columns

```
Booking_ID                            object
no_of_adults                           int64
no_of_children                         int64
no_of_weekend_nights                   int64
no_of_week_nights                      int64
type_of_meal_plan                     object
required_car_parking_space             int64
room_type_reserved                    object
lead_time                              int64
arrival_year                           int64
arrival_month                          int64
arrival_date                           int64
market_segment_type                   object
repeated_guest                         int64
no_of_previous_cancellations           int64
no_of_previous_bookings_not_canceled   int64
avg_price_per_room                   float64
no_of_special_requests                 int64
booking_status                        object
```

- The dataset contains a mix of numerical and categorical variables, which is suitable for prediction modeling.
- Numerical variables (int64 and float64) represent quantitative information such as:
  - Number of adults, children, nights stayed
  - Lead time of booking
  - Previous cancellations and booking history
  - Average price per room
  - Number of special requests
- Categorical variables (object type) include:
  - type_of_meal_plan
  - room_type_reserved
  - market_segment_type
  - booking_status
  - Booking_ID
- The Booking_ID column is an identifier and does not contribute to prediction; therefore, it can be removed during preprocessing.
- Since machine learning models require numerical inputs, categorical variables will need encoding (e.g., one-hot encoding) before model building.

## 4.3 Checking Statistical Summary of the Data

```
       no_of_adults  no_of_children  no_of_weekend_nights  no_of_week_nights  \
count  36275.000000    36275.000000          36275.000000       36275.000000
mean       1.844962        0.105279              0.810724           2.204300
std        0.518715        0.402648              0.870644           1.410905
min        0.000000        0.000000              0.000000           0.000000
25%        2.000000        0.000000              0.000000           1.000000
50%        2.000000        0.000000              1.000000           2.000000
75%        2.000000        0.000000              2.000000           3.000000
max        4.000000       10.000000              7.000000          17.000000

       required_car_parking_space     lead_time  arrival_year  arrival_month  \
count                36275.000000  36275.000000  36275.000000   36275.000000
mean                     0.030986     85.232557   2017.820427       7.423653
std                      0.173281     85.930817      0.383836       3.069894
min                      0.000000      0.000000   2017.000000       1.000000
25%                      0.000000     17.000000   2018.000000       5.000000
50%                      0.000000     57.000000   2018.000000       8.000000
75%                      0.000000    126.000000   2018.000000      10.000000
max                      1.000000    443.000000   2018.000000      12.000000

       arrival_date  repeated_guest  no_of_previous_cancellations  \
count  36275.000000    36275.000000                  36275.000000
mean      15.596995        0.025637                      0.023349
std        8.740447        0.158053                      0.368831
min        1.000000        0.000000                      0.000000
25%        8.000000        0.000000                      0.000000
50%       16.000000        0.000000                      0.000000
75%       23.000000        0.000000                      0.000000
max       31.000000        1.000000                     13.000000

       no_of_previous_bookings_not_canceled  avg_price_per_room  \
count                          36275.000000        36275.000000
mean                               0.153411          103.423539
std                                1.754171           35.089424
min                                0.000000            0.000000
25%                                0.000000           80.300000
50%                                0.000000           99.450000
75%                                0.000000          120.000000
max                               58.000000          540.000000

       no_of_special_requests
count             36275.000000
mean                  0.619655
std                   0.786236
min                   0.000000
25%                   0.000000
50%                   0.000000
75%                   1.000000
max                   5.000000
```

- The statistical summary provides key insights into the distribution and spread of numerical variables in the dataset.
- The average number of adults per booking is approximately 1.84, indicating that most bookings are made for 1-2 adults.
- The mean number of children is very low (0.10), showing that most bookings do not include children.
- The average number of weekend nights (0.81) and week nights (2.20) suggests that most guests stay for short durations, typically 2-3 nights.
- The variable **lead_time** has a high standard deviation (85.93) and a wide range (0 to 443), indicating high variability in how early customers book rooms.
- The **arrival_year** ranges between 2017 and 2018, showing that the dataset covers bookings from a limited time period.
- The **average price per room** is around 103, with values ranging from 0 to 540, indicating the presence of price variability and possible outliers.
- The variables **no_of_previous_cancellations** and **no_of_previous_bookings_not_canceled** have low mean values, suggesting that most customers do not frequently cancel bookings.
- This mean number of special requests is 0.62, indicating that most customers either make no special requests or only one request.
- The binary variable **required_car_parking_space** has a low mean (~0.03), showing that only a small proportion of customers require parking.

## 4.4 Checking for Data Irregularities & Treating them

### 4.4.1 Checking for duplicate records

- The dataset contains 0 duplicate records, indicating that each entry is unique.
- This ensures that there is no repeated booking information, which helps maintain the accuracy and reliability of the analysis.
- Since no duplicates are present, no data removal or cleaning is required for duplicate values.

### 4.4.2 Checking for logical inconsistencies

- 139 records show bookings with children but no adults, which may indicate data entry errors or unusual booking cases that may require verification.
- 78 records have zero total nights, meaning no stay duration was recorded. These observations may not be meaningful for analysis and may need to be removed or reviewed.
- 545 records have zero or negative average price per room, which is unrealistic since hotel rooms generally have a positive price. These values may represent data entry errors, promotional bookings, or missing price information.
- There are no records with negative lead time, indicating that booking dates are logically consistent.
- There are no cases where the number of special requests exceeds 5, suggesting that the special requests variable falls within a reasonable range.

### 4.4.3 Checking for invalid or unrealistic values

- The negative value check shows 0 negative values across key numerical variables such as **lead_time**, **avg_price_per_room**, **number of adults, children, and nights**, indicating that the dataset follows logical numerical constraints.
- The minimum values for most count-based variables (adults, children, nights, special requests) are 0, which is reasonable since some bookings may not include children, weekend stays, or special requests.
- The **lead_time** variable has a wide range from 0 to 443 days, indicating that customers book rooms both at the last minute and several months in advance.
- The average price per room ranges from 0 to 540, showing significant variation in room pricing, which may depend on room type, season, or demand.
- The number of weekend nights ranges from 0 to 7, while week nights range from 0 to 17, suggesting variation in stay duration among customers.

- The number of special requests ranges from 0 to 5, indicating that customers typically make a limited number of additional service requests.

### 4.4.4 Checking consistency of Categorical Data

- The categorical variables **type_of_meal_plan**, **room_type_reserved**, **market_segment_type**, and **booking_status** were checked for formatting inconsistencies such as extra spaces and inconsistent letter casing.
- After applying string cleaning operations (strip and lowercase), the categories were found to be consistent, indicating that there are no duplicate category labels caused by formatting issues.
- The meal plan feature contains four categories: *Mean Plan 1*, *Meal Plan 2*, *Meal Plan 3*, and *Not Selected*, with **Meal Plan 1** being the most frequently chosen option.
- The room type reserved variable shows that **Room_Type 1** is the most commonly booked room, while some room types such as **Room_Type 3** and **Room_Type 7** appear very rarely.
- The market segment type indicates that most bookings are made through the Online segment, followed by Offline and Corporate channels. Aviation and Complementary segments represent only a small proportion of bookings.
- The target variable (**booking_status**) shows that the dataset contains both canceled and non-canceled bookings, with **Not_Canceled** bookings being more frequent, indicating a moderately imbalanced dataset.

### 4.4.5 Treating unrealistic or inconsistent records

- Records where children were present but no adults were included were removed, as such cases are logically unrealistic in hotel bookings.
- Bookings with zero total nights were removed, since they do not represent actual stays and do not contribute meaningful information for analysis.
- Records with zero or negative average price per room were also removed, as room prices are expected to be positive and such values indicate possible data entry errors or invalid transactions.
- Removing these inconsistent records helps improve data quality and reliability, ensuring that the dataset better reflects real-world booking behavior.

## 4.5 Univariate Analysis

### 4.5.1 Identifying numerical and categorical variables

```
Numerical Variables:
Index(['no_of_adults', 'no_of_children', 'no_of_weekend_nights',
       'no_of_week_nights', 'required_car_parking_space', 'lead_time',
       'arrival_year', 'arrival_month', 'arrival_date', 'repeated_guest',
       'no_of_previous_cancellations', 'no_of_previous_bookings_not_canceled',
       'avg_price_per_room', 'no_of_special_requests'],
      dtype='object')

Categorical Variables:
Index(['Booking_ID', 'type_of_meal_plan', 'room_type_reserved',
       'market_segment_type', 'booking_status'],
      dtype='object')
```

- The dataset contains both numerical and categorical features, which are important for understanding booking patterns and building predictive models.
- A total of 13 numerical variables were identified, including features related to customer composition (number of adults, children), booking duration (weekend and week nights), booking behavior (lead time, previous cancellations), and pricing information (average price per room).
- These numerical variables help capture quantitative patterns in customer booking behavior.
- 5 categorical variables were identified: Booking_ID, type_of_meal_plan, room_type_reserved, market_segment_type, and booking_status.
- The column Booking_ID serves as a unique identifier and does not provide predictive value, so it can be removed during preprocessing.

- The remaining categorical variables represent customer preferences and booking channels, which need to be converted into numerical form using encoding techniques before applying machine learning models.

### 4.5.2 Exploring Numerical Variables:

**1. Number of Adults**

- The majority of bookings are made for 2 adults, indicating that most hotel reservations are for couples or pairs of travelers.
- Bookings with 1 adult are the second most common, representing solo travelers.
- A smaller proportion of bookings include 3 adults, while bookings with 4 adults are very rare.
- The distribution suggests that most hotel stays involve small groups rather than large groups.

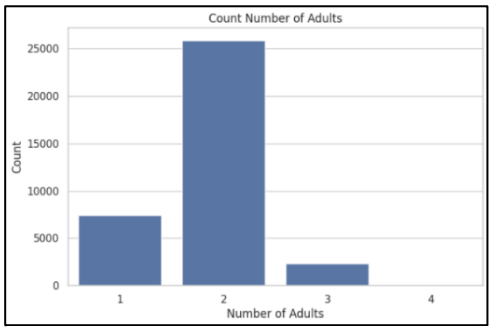


*Figure 1: Distribution plot for Number of Adults*

**2. Number of Children**

- The majority of bookings have 0 children, indicating that most guests travel without children.
- A small number of bookings include 1 or 2 children, suggesting that family bookings are less common compared to individual or couple bookings.
- Very few records show higher number of children, which may represent rare cases or potential outliers.
- The distribution is highly skewed toward zero, showing that the hotel is primarily used by adults traveling alone or in pairs rather than large families.

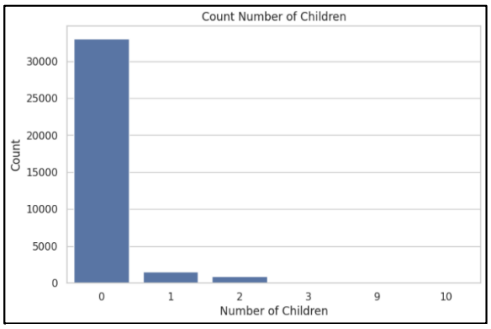


*Figure 2: Distribution plot for Number of Children*

**3. Number of Weekend Nights**

- A large number of bookings have 0 weekend nights, indicating that many guests do not stay during weekends, possibly due to business or weekday travel.
- Bookings with 1 or 2 weekend nights are also common, suggesting that some customers prefer short weekend stays or short trips.
- Very few bookings include 3 or more weekend nights, indicating that long weekend stays are relatively uncommon.
- The distribution shows that most customers prefer short-duration stays rather than extended weekend vacations.

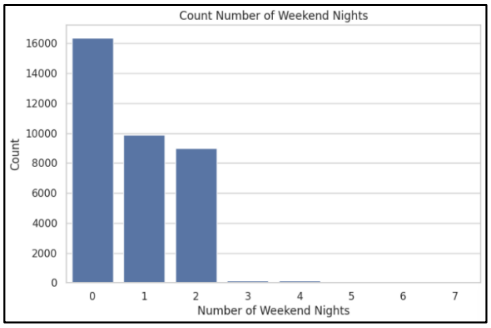


*Figure 3: Distribution plot for Number of Weekend Nights*

**4. Number of Week Nights**

- Most bookings include 1-3 week nights, indicating that customers typically stay for short durations during weekdays.
- The highest frequency is observed at 2 week nights, suggesting that short mid-week stays are common.
- The number of bookings gradually decreases as the number of week nights increases, showing that long weekday stays are relatively rare.
- Only a small number of bookings exceed 5 week nights, indicating that extended stays are uncommon.
- The distribution suggests that customers generally prefer short trips rather than long hotel stays during weekdays.

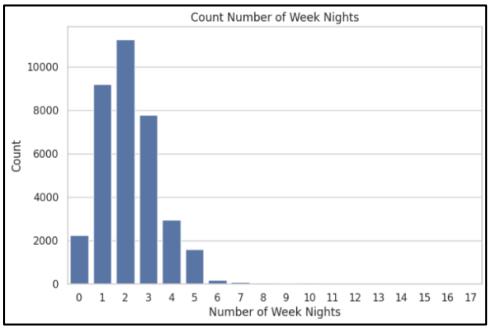


*Figure 4: Distribution plot for Number of Week Nights*

### 5. Lead Time

- The distribution of lead time is positively skewed (skewness = 1.28), indicating that most bookings are made closer to the arrival date, while fewer bookings are made far in advance.
- A large concentration of bookings occurs within 0-100 days before arrival, showing that customers typically prefer short to medium advance booking periods.
- The frequency gradually decreases as lead time increases, with relatively fewer bookings made more than 200 days in advance.
- A small number of bookings have very high lead times (above 300 days), which may be considered extreme values or outliers.
- The right-skewed distribution suggests that last-minute or moderately early bookings are more common than long-term advance bookings.

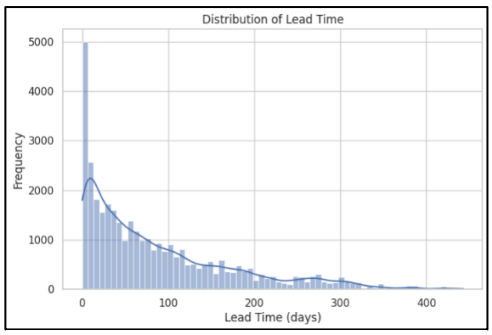


*Figure 5: Distribution plot for Lead Time*

### 6. Average Price per Room

- The distribution of average room price is positively skewed (skewness = 1.16), indicating that most room prices are concentrated in the lower to mid-price range, while fewer bookings have very high prices.
- The majority of room prices fall approximately between 80 and 120, suggesting that this is the most common pricing range for bookings.
- As the price increases beyond 150, the frequency of bookings gradually decreases, showing that higher-priced rooms are less frequently booked.
- A small number of bookings have very high prices (above 300), which may be considered outliers or premium room bookings.
- The right-skewed distribution suggests that customers are more likely to choose moderately priced rooms rather than expensive options.

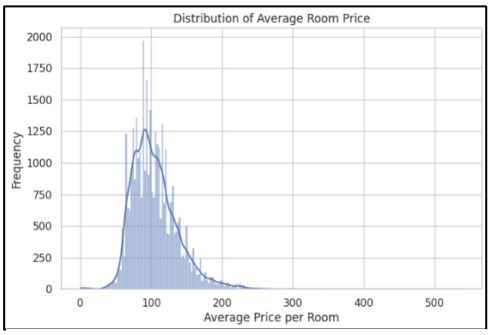


*Figure 6: Distribution plot for Average Price per Room*

### 7. Number of Special Requests

- The majority of bookings have 0 special requests, indicating that most customers do not request additional services beyond the standard booking.
- A significant number of customers make 1 special request, suggesting some preference for minor customizations such as room preferences or additional facilities.
- The number of bookings decreases as the number of special requests increases, showing that customers rarely make multiple special requests.
- Very few bookings contain 3 or more special requests, indicating that such cases are uncommon.
- The distribution suggests that most customers prefer simple bookings with minimal customization.

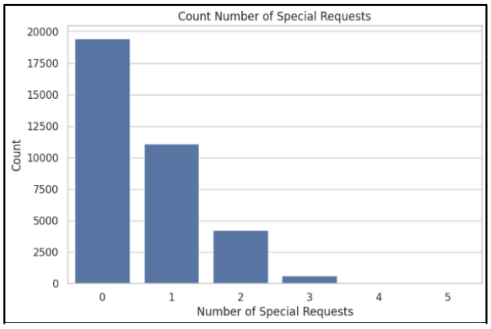


*Figure 7: Distribution plot for Number of Special Requests*

## 4.5.3 Exploring Categorical Variables:

### 1. Booking Status

- The dataset contains a higher number of Not_Canceled bookings compared to Canceled bookings, indicating that most customers complete their reservations successfully.
- However, the number of cancellations is still significant, showing that booking cancellation is an important issue that needs to be analyzed.
- The distribution shows a moderate class imbalance, where the number of non-canceled bookings is noticeably higher than canceled bookings.
- This imbalance may affect model performance, as the model may tend to predict the majority class more frequently.

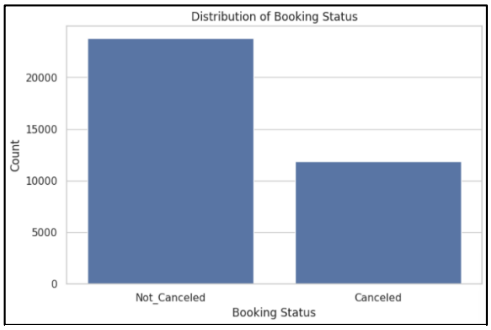


*Figure 8: Distribution plot for Booking Status*

## 2. Market Segment Type

- The majority of bookings come from the Online market segment, indicating that most customers prefer to book rooms through online platforms or websites.
- The Offline segment also contributes a significant number of bookings, showing that traditional booking channels are still relevant.
- The Corporate segment accounts for a smaller proportion of bookings, suggesting that business-related reservations are less frequent compared to individual bookings.
- The Aviation and Complementary segments represent very few bookings, indicating that these booking channels are rarely used.
- The distribution suggests that digital platforms play a major role in attracting customers, highlighting the importance of online booking systems.

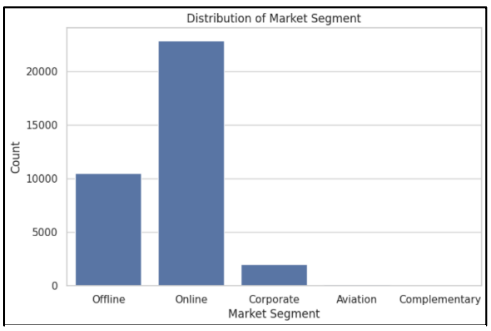


*Figure 9: Distribution plot for Market Segment*

## 3. Type of Meal Plan

- The majority of customers prefer Meal Plan 1, indicating that this is the most popular meal option among guests.
- A considerable number of customers have not selected any meal plan, suggesting that some guests may prefer flexibility in their dining choices or choose external food options.
- Meal Plan 2 is selected by a smaller group of customers, showing moderate preference.
- Meal Plan 3 has very few bookings, indicating that it is the least preferred meal option.
- The distribution shows that most customers opt for basic or standard meal plans, while premium or alternative meal options are less frequently selected.

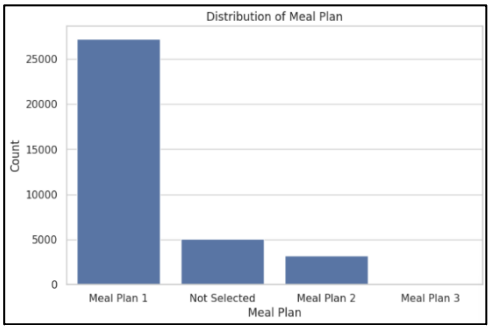


*Figure 10: Distribution plot for Type of Meal Plan*

## 4. Reserved Room Type

- The majority of bookings are for **Room_Type 1**, indicating that this room category is the most preferred among customers.
- **Room_Type 4** is the second most frequently booked option, but its count is significantly lower than Room_Type 1.
- Other room types such as **Room_Type 2**, **Room_Type 5**, **Room_Type 6**, **Room_Type 7**, and **Room_Type 3** have relatively very low booking frequencies.
- The distribution suggests that most customers prefer standard or commonly available room types, while premium or less common room categories are selected less frequently.
- The large difference in booking frequency between room types indicates uneven demand across room categories.

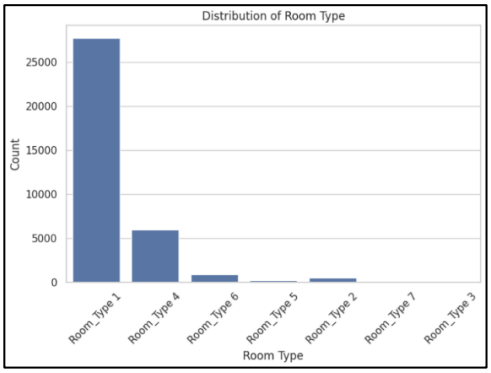


*Figure 11: Distribution plot for Reserved Room Type*

## 5. Repeated Guest

- The majority of bookings are from non-repeated guests (0), indicating that most customers are first-time visitors.
- Only a very small proportion of bookings come from repeated guests (1), showing low customer return frequency.
- The distribution is highly imbalanced, with significantly fewer repeat customers compared to new customers.
- This suggests that the hotel may have limited customer retention or repeat booking behavior.

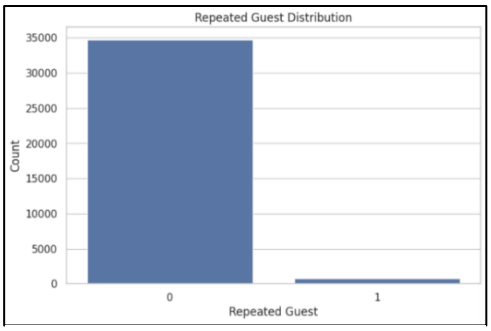


*Figure 12: Distribution plot for Repeated Guest*

### 6. Required Car Parking Space

- The majority of bookings do not require car parking space (0), indicating that most customers either do not travel by car do not need parking facilities.
- Only a very small proportion of bookings require car parking space (1), showing low demand for parking services.
- The distribution is highly imbalanced, with significantly fewer customers requesting parking compared to those who do not.
- This suggests that parking may not be a primary concern for most guests, or customers may rely on public transport, taxis, or other travel options.

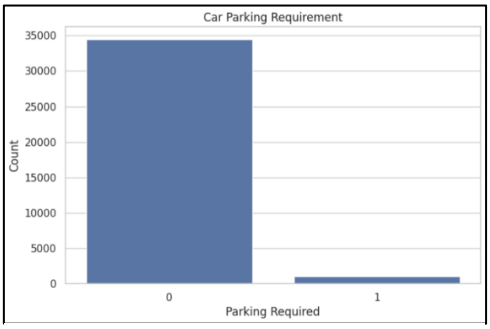


*Figure 13: Distribution plot for Required Car Parking Space*

### 7. Arrival Month

- The number of bookings varies across different months, indicating the presence of seasonal patterns in hotel demand.
- The highest number of bookings is observed in September and October, suggesting that these months may represent peak travel periods.
- Bookings gradually increase from the beginning of the year and reach a peak in the later months, showing higher demand during the second half of the year.
- The lowest number of bookings is observed in the early months such as January and February, indicating relatively lower travel activity during this period.
- The variation in bookings across months suggests that customer demand may be influenced by seasonal trends, holidays, or travel preferences.

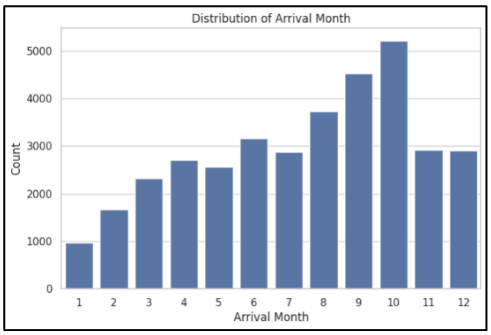


*Figure 15: Distribution plot for Month of Arrival*

## 4.6 Bivariate Analysis

### 4.6.1 Numerical Variables vs Target Variable **(booking_status)**

### 1. Lead Time vs Booking Status

- The average lead time is significantly higher for canceled bookings compared to non-canceled bookings.
- Customers who canceled their bookings tend to make reservations much earlier in advance than those who complete their stay.
- Non-canceled bookings generally have shorter lead times, indicating that customers who book closer to the check-in date are more likely to honor their reservations.
- This suggests that longer lead times may increase the probability of cancellation, possibly because customers' plans change over time.

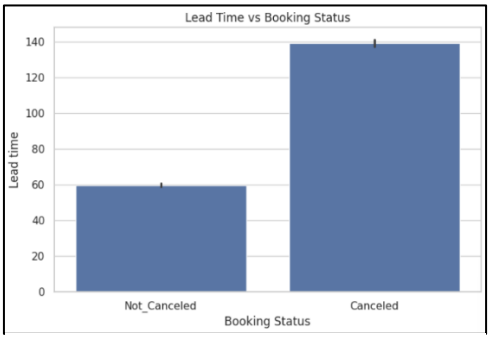


*Figure 16: Lead time vs Booking Status*

### 2. Average Price per Room vs Booking Status

- The average price per room is slightly higher for canceled bookings compared to non-canceled bookings.
- This suggests that customers who book higher-priced rooms may be more likely to cancel their reservations.
- Non-canceled bookings tend to have a slightly lower average room price, indicating that moderately priced rooms may have more stable confirmations.
- The difference in average price is not extremely large, but it indicates that price may have some influence on cancellation behavior.

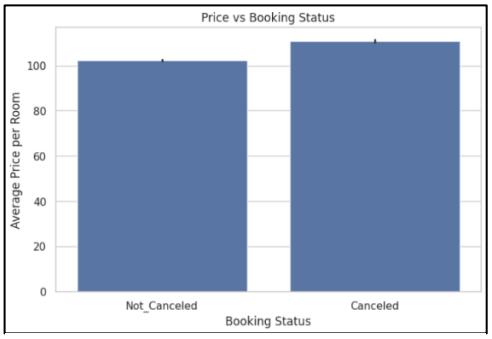


*Figure 17: Average Price per Room vs Booking Status*

### 3. Week Nights vs Booking Status

- The average number of week nights is slightly higher for canceled bookings compared to non-canceled bookings.
- This indicates that customers who plan longer weekday stays may have a slightly higher tendency to cancel their reservations.
- Non-canceled bookings generally involve slightly shorter weekday stays, suggesting that shorter trips may have a higher confirmation rate.
- However, the difference between the two groups is relatively small, indicating that the number of week nights alone may not strongly influence cancellation behavior.

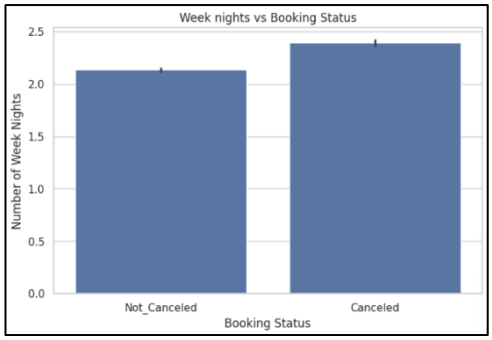



*Figure 18: Week Nights vs Booking Status*

### 4. Weekend Nights vs Booking Status

- The average number of weekend nights is slightly higher for canceled bookings compared to non-canceled bookings.
- Customers who planned longer weekend stays appear to have a slightly higher likelihood of canceling their reservations.
- Non-canceled bookings generally involve slightly fewer weekend nights, indicating the shorter weekend stays may have a higher confirmation rate.
- However, the difference between the two categories is relatively small, suggesting that weekend nights alone may not strongly influence cancellation behavior.

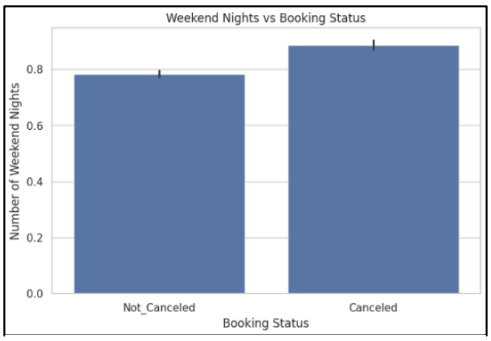



*Figure 19: Weekend Nights vs Booking Status*

### 5. Number of Special Requests vs Booking Status

- The average number of special requests is higher for non-canceled bookings compared to canceled bookings.
- Customers who make more special requests appear to be more certain about their stay, indicating stronger booking commitment.
- Canceled bookings generally show fewer special requests, suggesting lower engagement or less planning certainty.
- This indicates that the number of special requests may have an inverse relationship with cancellation probability.

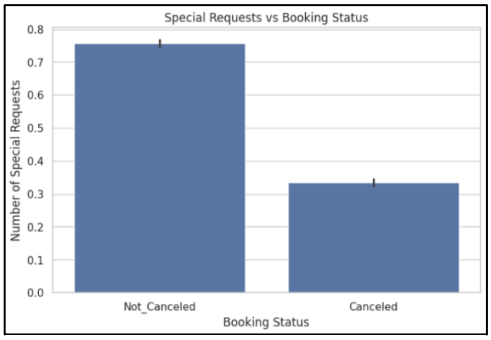



*Figure 20: Number of Special Requests vs Booking Status*

### 6. Number of Adults vs Booking Status

- The average number of adults is slightly higher for canceled bookings compared to non-canceled bookings.
- However, the difference between the two groups is very small, indicating that number of adults does not strongly influence cancellation behavior.
- Most bookings in both categories involve approximately 2 adults, suggesting that couples or small groups are the most common type of guests.
- The similarity in values for both categories indicates that this feature alone may have limited impact on predicting cancellations.

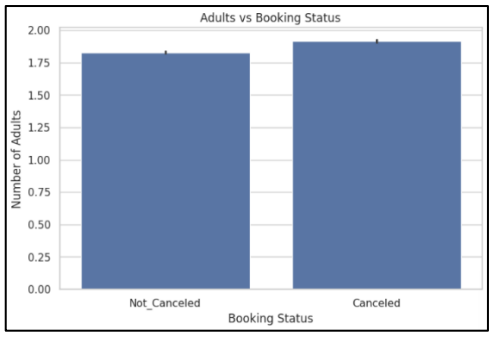



*Figure 21: Number of Adults vs Booking Status*

### 7. Number of Children vs Booking Status

- The average number of children is slightly higher in canceled bookings compared to non-canceled bookings.
- This suggests that bookings involving children may have a marginally higher chance of cancellation, possibly due to uncertainties in family travel plans.
- However, the overall values are quite low (close to 0 for both groups), indicating that most bookings do not include children.
- Since the difference between the two groups is small, this variable may have moderate but not strong predictive importance in determining cancellations.

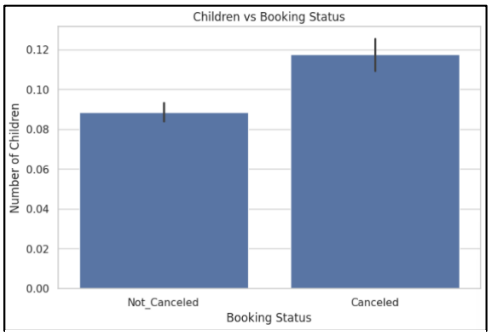


*Figure 22: Number of Children vs Booking Status*

## 4.6.2 Categorical Variables vs Target Variable **(booking_status)**

### 1. Market Segment Type vs Booking Status

- Online Segment has the highest number of bookings as well as cancellations, indicating that most customers book through online platforms.
- A significant portion of cancellations also comes from the online market segment, suggesting that online bookings may be more prone to cancellation due to ease of booking and flexible decision-making.
- The offline segment also shows a considerable number of bookings, but cancellations are relatively lower compared to online bookings.
- Corporate bookings show comparatively fewer cancellations, indicating that business-related bookings tend to be more stable and planned.
- Aviation and complementary segments have very low booking volumes, suggesting they contribute minimally to overall cancellation patterns.

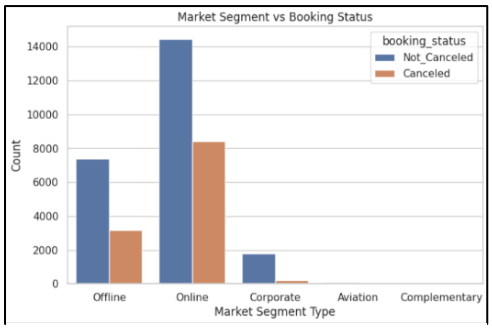


*Figure 23: Market Segment Type vs Booking Status*

### 2. Meal Plan Type vs Booking Status

- Meal Plan 1 has the highest number of bookings as well as cancellations, indicating that it is the most commonly chosen meal option among customers.
- Although cancellations are present in Meal Plan 1, the number of non-canceled bookings is significantly higher, suggesting that this meal plan is generally reliable.
- Customers who selected "Not Selected" meal option also show a noticeable number of cancellations, which may indicate lower commitment or flexibility in booking preferences.
- Meal Plan 2 has relatively fewer bookings but shows a moderate proportion of cancellations, suggesting slightly higher uncertainty among customers choosing this plan.
- Meal Plan 3 has extremely low frequency, indicating it is rarely selected and has minimal influence on overall cancellation patterns.

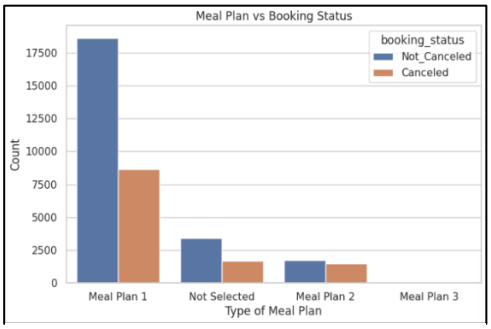


*Figure 24: Meal Plan Type vs Booking Status*

### 3. Reserved Room Type vs Booking Status

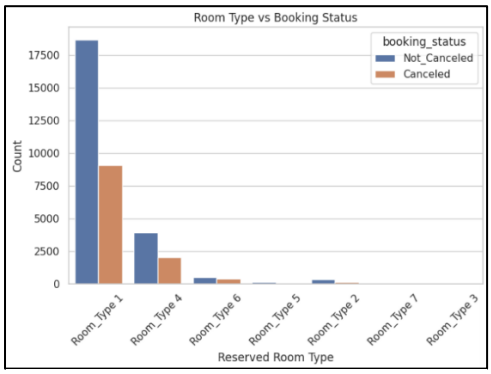


*Figure 25: Reserved Room Type vs Booking Status*

- Room Type 1 has the highest number of bookings as well as cancellations, indicating that it is the most preferred room category among customers.
- Even though cancellations are present in Room Type 1, the number of successful (non-canceled) bookings is much higher, showing strong customer preference and reliability for this room category.
- Room Type 4 is the second most frequently booked room type and also shows a moderate number of cancellations.
- Other room types such as Room Type 2, 3, 5, 6, and 7 have very low booking counts, indicating limited popularity and minimal impact on overall cancellations trends.
- Since the majority of bookings are concentrated in only a few room categories, room type may still play a role in cancellation prediction, but its effect appears less significant compared to variables like lead time or special requests.

### 4. Repeated Guest vs Booking Status

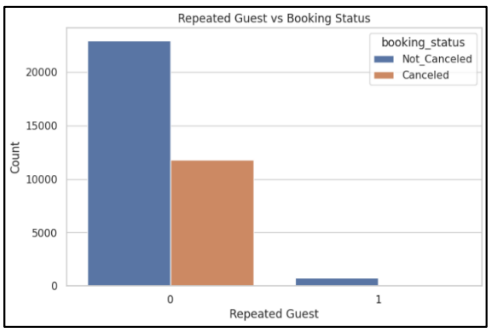


*Figure 26: Repeated Guest vs Booking Status*

- The majority of bookings come from new customers (not repeated guests), indicating that most reservations are made by first-time visitors.
- Among non-repeated guests, a noticeable number of cancellations is observed, suggesting that new customers are more likely to cancel bookings.
- Repeated guests show very low cancellation counts, indicating that returning customers are generally more reliable and committed to their bookings.
- This suggests that customer loyalty plays an important role, as guests who have previously stayed at the hotel are les likely to cancel compared to first-time customers.

### 5. Required Car Parking Space vs Booking Status

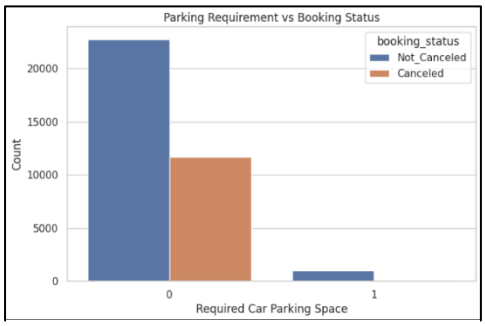


*Figure 27: Required Car Parking Space vs Booking Status*

- The majority of customers do not require a car parking space, indicating that most guests either do not travel by personal vehicle or parking is not a primary concern.
- Among customers who did not request parking, both non-canceled and canceled bookings are observed, suggesting that lack of parking requirement does not strongly influence cancellation behavior.
- Customers who required parking space represent a very small portion of total bookings, and cancellation behavior.
- The low frequency of parking requests indicates that this variable may have limited impact on predicting booking cancellations.

### 6. Arrival Month vs Booking Status

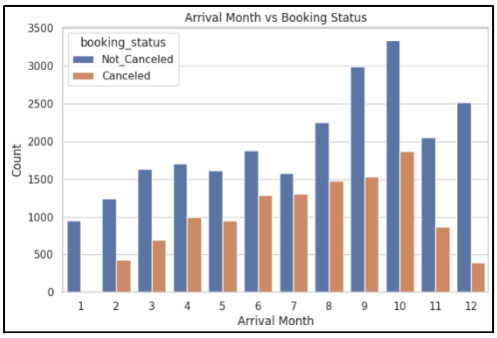


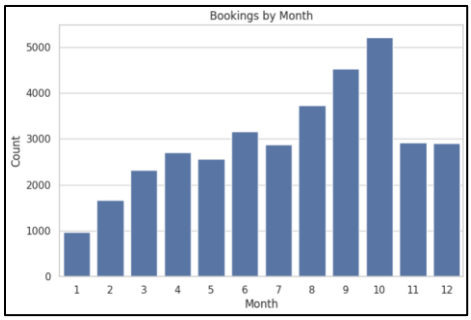
*Figure 28: Arrival Month vs Booking Status*

- Bookings are not evenly distributed across months, indicating seasonal variation in hotel demand.
- The highest number of bookings is observed in August, September, and October, suggesting these are peak travel periods.
- Cancellations also increase during high-demand months, especially around September and October, which may indicate greater uncertainty during peak seasons.
- Early months such as January and February show comparatively lower booking volumes as well as fewer cancellations.

- The gap between non-canceled and canceled bookings is visible across all months, but cancellations appear relatively higher during peak season months, possibly due to advance bookings and changing travel plans.

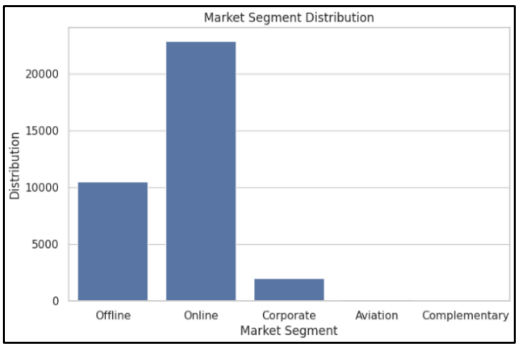
## 4.7 Answers to the EDA Questions

### 4.7.1 What are the busiest months in the hotel?



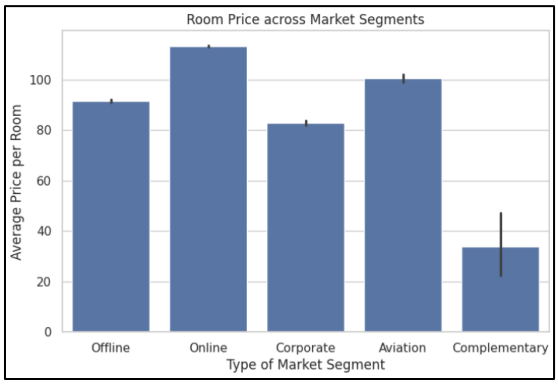

The hotel experiences the highest number of bookings during October, making it the busiest month overall. Booking volume also remains high in September and August, indicating that the period from late summer to early autumn represents the peak season for the hotel. There is a noticeable upward trend in bookings starting from June, which continues to rise until it reaches its maximum in October. In contrast, the months travel activity during the beginning of the year. Overall, the data indicates a clear seasonal pattern in hotel demand, with significantly higher customer activity during the mid to later months of the year.

### 4.7.2 Which market segment do the most of the guests come from?




Most of the guests come from the Online market segment, which shows the highest number of bookings compared to all other segments. The Offline segment is the second largest contributor, but its count is significantly lower than Online bookings. The Corporate segment accounts for a much smaller proportion of guests, while Aviation and Complementary segments contribute very minimally to the total bookings. This indicates that the majority of customers prefer making reservations through online platforms, highlighting the importance of digital channels in attracting hotel bookings.

### 4.7.3 How rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?




Room prices vary noticeably across different market segments, indicating that pricing is dynamic and influenced by customer type and demand patterns. The Online segment shows the highest average room price, suggesting that customers booking through online platforms may be less price-sensitive or more willing to pay for convenience and flexibility. The Aviation segment also has relatively high room prices, which may be associated with urgent or last-minute travel needs. The Offline segment shows moderately high pricing, indicating stable demand through traditional booking channels.

In contrast, the Corporate segment has comparatively lower average room prices, possibly due to negotiated corporate rates or long-term business partnerships that provide discounted pricing. The Complementary segment shows the lowest room price, which is expected since these bookings may include promotional stays, partnerships, or complementary services offered by the hotel.

Overall, the differences in pricing across market segments demonstrates that hotel room rates are influenced by demand patterns, booking channels, and customer profiles. Segments with higher demand or urgency tend to have higher room prices, while segments involving business agreements or promotional arrangements tend to have lower pricing.

### 4.7.4 What percentage of bookings are cancelled?

```
booking_status
Not_Canceled    66.750948
Canceled        33.249052
```

Approximately 33.25% of the bookings are cancelled.

This indicates that about 1 out of every 3 bookings does not result in an actual stay, which is a significant proportion. Such a high cancellation rate can impact revenue forecasting, room availability planning, and operational efficiency. It highlights the importance of developing predictive models to identify potential cancellations in advance so that the hotel can take proactive actions such as overbooking strategies, targeted reminders, or flexible pricing policies.

### 4.7.5 Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?

```
booking_status          0          1
repeated_guest
0               66.029547  33.970453
1               98.007472   1.992528
```

Among the repeating guests, approximately 1.99% of bookings are cancelled, while about 98.01% of bookings are not canceled. This indicates that repeat guests are highly reliable customers and rarely cancel their bookings. Since these guests are already familiar with the hotel's services and have chosen to return, they demonstrate stronger trust and loyalty towards the brand. The extremely low cancellation percentage highlights the importance of maintaining strong relationships with repeat customers, as they contribute significantly to stable revenue and brand equity.

### 4.7.6 Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?

Yes, special requirements appear to affect the cancellation of bookings. The chart shows that guests who make **no special requests (0 requests)** have the highest number of cancellations compared to those who specify one or more requests. As the number of special requests increases from 1 to 3, the number of cancellations decreases significantly, while the proportion of non-canceled bookings remains relatively high.

This suggests that guests who provide special requests are generally **more certain about their stay and more committed to the booking**, reducing the likelihood of cancellation. On the other hand, guests who do not specify any special requirements may be less engaged in the booking process and therefore more likely to cancel.

Overall, the analysis indicates that the number of special requests is inversely related to the cancellation probability, such that bookings with more special requests are less likely to be canceled.

## 4.8 Key Insights

### 4.8.1 Data Structure & Quality

- The dataset contains 36,275 records and 19 variables, providing sufficient observations for reliable modeling.
- Variables include both numerical features (lead time, price, number of nights, etc) and categorical features (meal plan, market segment, room type, etc).
- No missing values are present in the dataset, indicating good data quality.
- No duplicate records were found, ensuring data consistency.
- Some logically inconsistent records were identified (e.g., children without adults, zero-night stays, zero room price) and were removed to improve dataset reliability.
- Categorical variables were checked for case sensitivity and extra spaces, ensuring consistent category representation.

### 4.8.2 Customer & Booking Characteristics (Univariate Analysis)

- Most bookings include 2 adults, indicating couples or small groups are the most common customers.
- The majority of bookings have 0 children, meaning most guests travel without children.
- Most customers stay 1-3 week nights, suggesting short-duration stays are common.
- Weekend stays are typically 0-2 nights, indicating limited weekend-only bookings.
- Most customers make 0-1 special requests, suggesting moderate customization needs.
- A very small percentage of customers are repeat guests, indicating most bookings come from new customers.
- Most customers do not require car parking, suggesting parking demand is relatively low.

### 4.8.3 Price & Lead Insights

- The average room price is concentrated around moderate price ranges, with some high-value outliers.
- Room price distribution is positively skewed, meaning most customers book affordable rooms while fewer customers book premium-priced rooms.
- Lead time is also positively skewed, indicating that most customers book relatively close to the arrival date.
- Some customers book rooms very far in advance, which may increase uncertainty and cancellation likelihood.

### 4.8.4 Cancellation Behavior Insights

- Around 33% of bookings are cancelled, which is a considerable proportion affecting revenue predictability.
- Majority of bookings (about 67%) are completed.
- Repeated guests show extremely low cancellation rates (~2%), indicating strong customer loyalty.
- Customers who make special requests are less likely to cancel, suggesting higher commitment to the booking.
- Bookings involving children show slightly higher cancellation tendency, possibly due to uncertainty in travel planning.

### 4.8.5 Market Segment Insights

- Most bookings come from the Online market segment, highlighting the importance of online booking platforms.
- The Offline segment is the second largest contributor to bookings.
- Corporate bookings represent a smaller but relatively stable customer segment.
- Aviation and complementary segments contribute very few bookings.
- Room prices very across market segments, showing that pricing strategies differ depending on booking channel and customer type.

### 4.8.6 Seasonal Trends

- Booking demand varies across months, showing clear seasonal patterns.
- The busiest months are August, September, and October, with October having the highest bookings.
- Early months such as January and February show lower booking volumes.
- Higher booking volumes during peak months also correspond to higher cancellations.

### 4.8.7 Bivariate Analysis Insights

**1. Lead Time vs Booking Status**

- Customers with longer lead times tend to cancel more frequently.
- Bookings are associated with higher uncertainty and change in travel plans.

**2. Price vs Booking Status**

- Bookings with higher average room prices show slightly higher cancellation tendency.
- Higher financial commitment may influence cancellation decisions.

**3. Week Nights vs Booking Status**

- Customers staying for more week nights show slightly higher cancellation probability.
- Longer stays may involve higher uncertainty or planning changes.

**4. Weekend Nights vs Booking Status**

- Customers with more weekend nights show a slightly higher likelihood of cancellation.
- Weekend plans may be more flexible or subject to change.

**5. Special Requests vs Booking Status**

- Customers who make more special requests are less likely to cancel.
- Higher engagement with booking preferences indicates stronger commitment.

**6. Number of Adults vs Booking Status**

- Number of adults shows minimal difference between cancelled and non-cancelled bookings.
- This variable alone is not a strong predictor of cancellation.

**7. Number of Children vs Booking Status**

- Bookings including children show slightly higher cancellation tendency, possibly due to uncertainties in family travel.

**8. Market Segment vs Booking Status**

- Online segment contributes the highest number of cancellations, mainly due to higher booking volume.
- Corporate bookings show relatively lower cancellation rates, indicating stable business travel demand.

**9. Meal Plan vs Booking Status**

- Most bookings are associated with Meal Plan 1, which also contributes most cancellations due to higher volume.
- Customers who do not select a meal plan show slightly higher cancellation tendency.

**10. Room Type vs Booking Status**

- Room Type 1 dominates bookings and cancellations.
- Other room types have limited representation, suggesting room category has moderate influence on cancellation.

**11. Arrival Month vs Booking Status**

- Cancellation patterns vary across months, with higher cancellations observed during peak booking months.
- Seasonal demand plays an important role in booking stability.

### 4.8.8 Overall EDA Conclusion

- Booking cancellations are influenced by a combination of customer behavior, booking characteristics, pricing, and seasonal trends.
- Important predictors of cancellation include:
  - Lead time
  - Special requests
  - Repeated guest status
  - Market segment
  - Price
  - Seasonal demand
- Customer commitment indicators such as repeat bookings and special requests reduce cancellation likelihood.
- The dataset shows strong potential for predictive modeling to help hotels anticipate cancellations and improve revenue management strategies.

# 5. Data processing

## 5.1 Outlier Detection

### 5.1.1 Visual Inspection of Outliers using Boxplot

**1. Number of Adults**

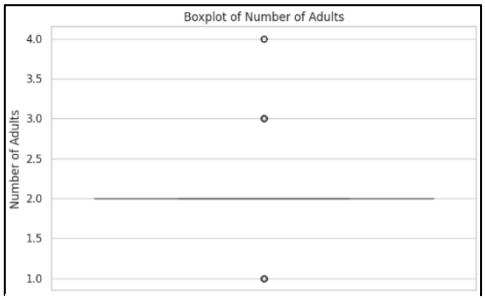


*Figure 29: Boxplot for 'no_of_adults.'*

- The median number of adults per booking is 2, indicating that most reservations are made for two people.
- The interquartile range (IQR) is very narrow, showing that the majority of bookings consistently involve 2 adults, with very little variation.
- A few outliers are present at 1, 3, and 4 adults, indicating that solo travelers and slightly larger groups exist but are relatively uncommon.
- The distribution suggests that couple or two-person bookings dominate the dataset, which aligns with typical hotel booking patterns.
- Since variation in this variable is limited, the number of adults may have a lower impact as a strong predictive feature for cancellation compared to variables like lead time or price.

**2. Number of Children**

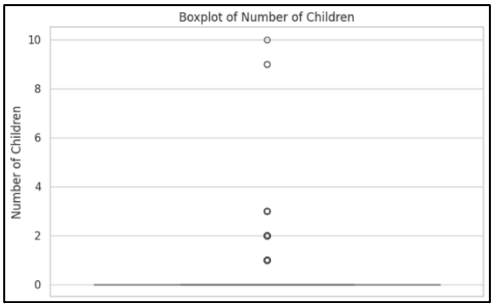


*Figure 30: Boxplot for 'no_of_children.'*

- The median value for number of children is 0, indicating that most bookings do not include children.
- The interquartile range (IQR) is concentrated at 0, showing that the majority of guests travel without children.
- Several outliers are present, with values ranging from 1 to 10 children, indicating that a few bookings involve large families or group reservations.
- The presence of extreme values such as 9 or 10 children suggests rare cases that may represent family events or data irregularities.
- Overall, the variable is highly skewed toward zero, meaning children are not a common component in most bookings.
- Since most observations are concentrated at zero, the number of children may have a limited influence individually, but could still contribute moderately to predicting cancellations when combined with other features.

**3. Number of Weekend Nights**

*Figure 31: Boxplot for 'no_of_weekend_nights.'*

- The median number of weekend nights is 1, indicating that most customers stay for only one weekend night.
- The interquartile range (IQR) lies between 0 and 2 nights, showing that the majority of bookings include short weekend stays.
- Many bookings have 0 weekend nights, suggesting that several customers stay only during weekdays.
- A few outliers are present at 6 and 7 weekend nights, indicating rare cases of extended weekend stays or long holiday bookings.
- The distribution is slightly right-skewed, meaning longer weekend stays are less common.
- Overall, customers generally prefer short weekend stays, and very long weekend bookings are uncommon.

## 4. Number of Week Nights

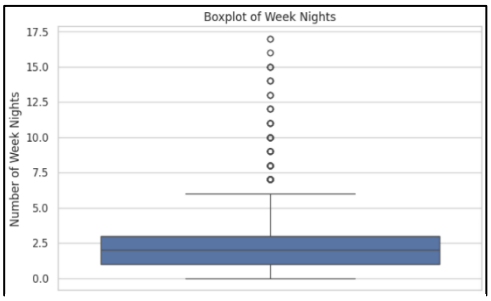


*Figure 32: Boxplot for 'no_of_week_nights.'*

- The median number of week nights is 2, indicating that most customers stay for around two weekdays.
- The interquartile range (IQR) lies approximately between 1 and 3 nights, showing that the majority of bookings are short weekday stays.
- Some bookings have 0 week nights, indicating purely weekend stays.
- A significant number of outliers are present beyond 6 nights, extending up to around 17 nights, representing long-duration stays.
- The distribution is right-skewed, meaning long weekday stays are less frequent but still occur.
- Overall, most customers prefer short weekday stays, while long stays are relatively rare but important.

## 5. Required Car Parking Space

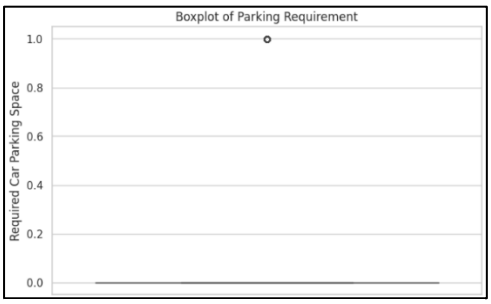


*Figure 33: Boxplot for 'required_car_parking_space.'*

- The median value is 0, indicating that the majority of customers do not require car parking space.
- The distribution is highly concentrated at 0, showing that parking is not a common requirement among most guests.
- A few values appear at 1, indicating that only a small proportion of customers request parking facilities.
- Since this variable is binary (0 = No parking required, 1 = Parking required), the boxplot shows very limited variation.
- The low frequency of parking requests suggests that parking availability may not be a major deciding factor for most customers when booking.
- Therefore, this variable may have relatively low influence on cancellation prediction, though it may still contribute when combined with other features.

## 6. Lead Time

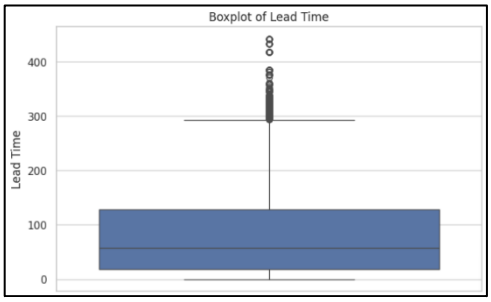


*Figure 34: Boxplot for 'lead_time'*

- The median lead time is around 60 days, indicating that most customers book their stay approximately 2 months in advance.
- The interquartile range (IQR) lies roughly between 20 and 125 days, showing that the majority of bookings are made within this time period.
- A large number of outliers are present, extending up to 400+ days, indicating that some customers book very far in advance.
- The distribution is positively skewed (right-skewed), meaning most bookings occur closer to the arrival date, while fewer bookings are made extremely early.
- The presence of many high-value outliers suggests that lead time varies significantly across customers, reflecting differences in travel planning behavior.
- Since higher lead times are often associated with greater uncertainty, this variable may play an important role in predicting booking cancellations.

## 7. Arrival Year

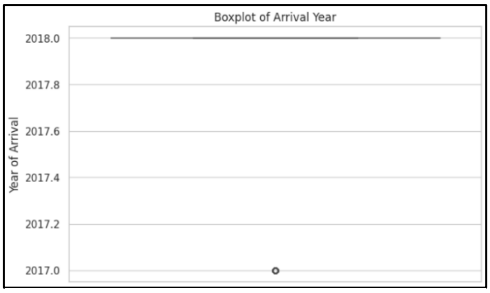


*Figure 35: Boxplot for 'arrival_year'*

- The variable arrival_year contains only two values: 2017 and 2018, indicating that the dataset covers bookings made within these two years.
- The majority of bookings appear to be concentrated in 2018, as the box is positioned near 2018.
- A few observations belong to 2017, which appear as separate points.
- Since this variable has very limited variation, it does not provide much distributional insight through a boxplot.
- The variable may still be useful to capture time-based booking patterns, but by itself it may not strongly influence cancellation prediction.

- Because only two years are present, this feature mainly helps in identifying temporal trends rather than variability.

## 8. Arrival Month

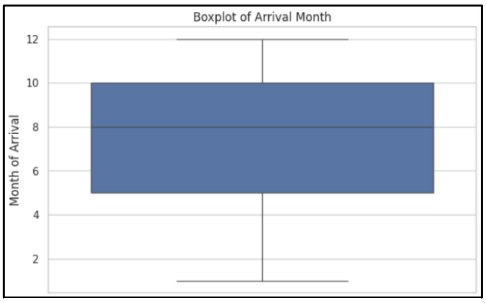


*Figure 36: Boxplot for 'arrival_month'*

- The median arrival month is around 8 (August), indicating that a large number of bookings occur in the later part of the year.
- The interquartile range (IQR) lies approximately between May (5) and October (10), showing that most bookings are concentrated in mid to late months of the year.
- The spread of data across months suggests seasonal booking behavior, with higher customer activity during peak travel periods.
- The whiskers extend from month 1 (January) to month 12 (December), indicating that bookings are distributed throughout the year.
- No significant outliers are observed, meaning booking months fall within the expected calendar range (1–12) and do not contain extreme or invalid values.
- Since arrival month represents a cyclical time variable, the absence of outliers confirms that the data is consistent and logically valid.

## 9. Repeated Guest

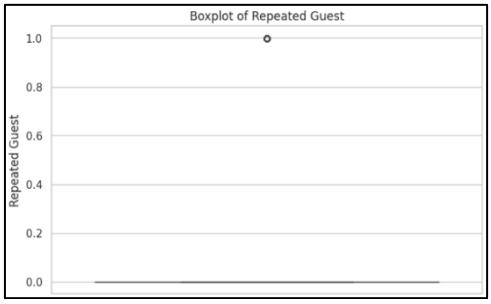


*Figure 37: Boxplot for 'repeated_guest'*

- The median value is 0, indicating that the majority of customers are first-time guests rather than repeat customers.
- The interquartile range (IQR) is concentrated at 0, showing that most observations belong to the non-repeated guest category.
- A few observations appear at value 1, which are visually identified as potential outliers in the boxplot.
- Since Repeated Guest is a binary categorical variable (0 or 1), these values are not true statistical outliers, but represent the minority class of repeat customers.
- No extreme or invalid values are observed, indicating the data is consistent, logically valid, and suitable for predictive modeling, although it suggests a class imbalance between new and returning customers.

## 10. Number of Previous Cancellations

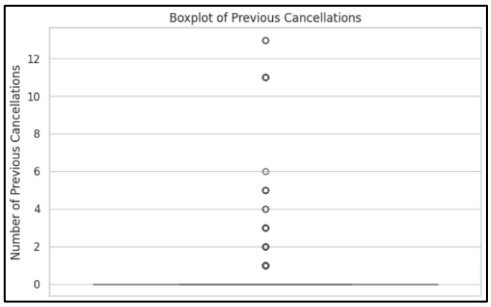


*Figure 38: Boxplot for 'no_of_previous_cancellations.'*

- The median value is 0, indicating that the majority of customers have not made any previous cancellations.
- The interquartile range (IQR) is concentrated at 0, showing that most observations lie in the lower range of the variable.
- Several data points appear above the upper whisker, indicating the presence of potential outliers, with some customers having a relatively high number of previous cancellations.
- These outliers suggest that a small group of customers frequently cancel bookings, which may indicate inconsistent booking behavior or lower commitment levels.
- Since the values are logically possible and relevant for predicting customer behavior, these outliers should not be removed, as they may significantly influence the predictive power of the classification model.

### 11. Number of Previous Non-Canceled Bookings

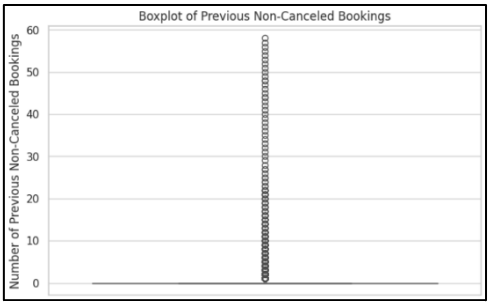


*Figure 39: Boxplot for 'no_of_previous_bookings_not_canceled.'*

- The median value is 0, indicating that the majority of customers do not have prior successful (non-canceled) bookings.
- The interquartile range (IQR) is concentrated at 0, showing that most observations lie in the lower range of the variable.
- A large number of data points appear above the upper whisker, indicating the presence of multiple potential outliers, with some customers having a high number of previous successful bookings.
- These outliers suggest that a small group of loyal or frequent customers repeatedly make bookings without canceling, reflecting strong customer retention behavior.
- Since these values are valid and meaningful for understanding customer loyalty patterns, the outliers should be retained, as they may significantly contribute to improving the predictive performance of the classification model.

### 12. Average Price per Room

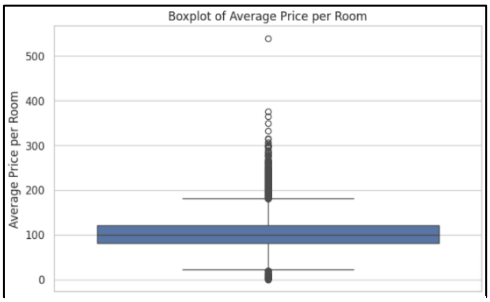


*Figure 40: Boxplot for 'avg_price_per_room.'*

- The median price lies around 100, indicating that the typical room price is concentrated near this value.
- The interquartile range (IQR) lies approximately between 80 and 120, showing that most room prices fall within a moderate price range.
- Several data points appear above the upper whisker, indicating the presence of high-value outliers, where some rooms are priced significantly higher than the typical range.
- A few observations are also seen below the lower whisker, representing low-price outliers, possibly due to discounts, promotional offers, or budget room categories.
- These outliers reflect variation in pricing strategies, such as premium room categories or seasonal price fluctuations, and should be retained as they provide meaningful information for predictive modeling.

### 13. Number of Special Requests

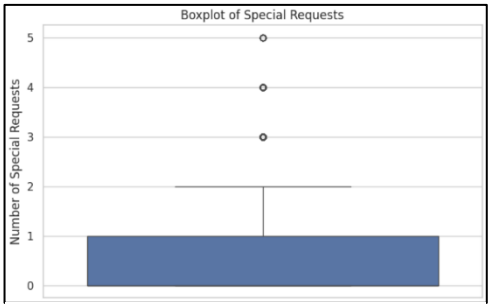


*Figure 41: Boxplot for 'no_of_special_requests.'*

- The median value is 0, indicating that most customers do not make any special requests during booking.
- The interquartile range (IQR) lies approximately between 0 and 1, showing that the majority of customers make very few special requests.
- A few observations appear above the upper whisker (values around 3, 4, and 5), indicating the presence of potential outliers with a higher number of special requests.
- These outliers suggest that a small group of customers has specific preferences or additional service requirements, reflecting more customized booking behavior.
- Since these values are valid and provide meaningful insight into customer needs and service expectations, the outliers should be retained as they may help improve the predictive performance of the classification model.

## 5.1.2 Detecting Outliers Using the IQR Method

```
lead_time: 1182 outliers detected
avg_price_per_room: 1090 outliers detected
no_of_weekend_nights: 21 outliers detected
no_of_week_nights: 319 outliers detected
no_of_special_requests: 721 outliers detected
```

- Outlier detection analysis indicates that **lead_time** contains 1182 outliers, indicating that some customers book rooms either far in advance or at very short notice, reflecting variation in booking-planning behavior.

- The variable **avg_price_per_room** exhibits 1090 outliers, indicating considerable variation in pricing attributable to factors such as room category, seasonal demand, promotional offers, or premium services.
- **no_of_weekend_nights** shows 21 outliers, indicating that only a small proportion of customers have unusually long weekend stays, while most bookings fall within the normal range.
- **No_of_week_nights** contains 319 outliers, suggesting that some customers opt for extended weekend stays, possibly related to business travel or long-duration visits.
- The variable **no_of_special_requests** has 721 outliers, indicating that certain customers make multiple service-related requests, reflecting diverse customer preferences and expectations.
- Since a considerable number of outliers are present across multiple variables, outlier treatment is required to reduce the impact of extreme values and improve model stability and predictive performance.

## 5.2 Treating Outliers

```
lead_time: 0 outliers remaining
avg_price_per_room: 0 outliers remaining
no_of_week_nights: 0 outliers remaining
no_of_special_requests: 0 outliers remaining
```

- Outliers in **lead_time**, **avg_price_per_room**, **no_of_week_nights**, and **no_of_special_requests** were treated using the Interquartile Range (IQR) method, which caps extreme values within the acceptable range.
- The lower and upper bounds were calculated using **Q1 – 1.5*IQR** and **Q3 + 1.5*IQR**, ensuring that extreme values were limited without removing any observations from the dataset.
- The clipping technique was applied to adjust extreme values to the calculated boundary limits, thereby reducing the impact of unusually high or low values.
- After treatment, 0 outliers remain in all selected variables, confirming that the dataset no longer contains extreme values beyond the acceptable range.
- This treatment helps in improving model stability; extreme values can negatively influence the performance of classification algorithms.
- Overall, the dataset is now more balanced and suitable for predictive modeling, ensuring more reliable and accurate results.

## 5.3 Feature Engineering

- The column **Booking_ID** was removed from the dataset as it serves only as a unique identifier and does not contribute meaningful information for predictive modeling.
- The target variable **booking_status** was converted into a binary format (0 and 1), where numerical encoding makes it suitable for classification algorithms.
- The value **0** represents **Not_Canceled** and **1** represents **Canceled**, ensuring a clear distinction between the two outcome classes.
- Categorical variables such as **type_of_meal_plan**, **room_type_reserved**, and **market_segment_type** were transformed into a numerical format using one-hot encoding, enabling the model to interpret these variables effectively.
- The parameter **drop_first = True** was applied to avoid the dummy variable trap and reduce multicollinearity among the generated variables.
- After preprocessing, the dataset is fully numeric, structured, and suitable for machine learning algorithms, ensuring improved model performance and interpretability.

## 5.4 Train – Test Split

- The dataset was divided into independent variables (X) and the target variable (Y), where booking_status is used as the dependent variable for prediction.
- The feature set X contains all input variables influencing booking behavior, while Y represents the output variable indicating whether the booking was canceled or not canceled.
- The data was split into training and testing sets using the **train_test_split()** function to evaluate model performance on unseen data.
- 80% of the data was allocated for training the model, allowing the algorithm to learn patterns and relationships among variables.

- The remaining 20% of the data was reserved for testing, ensuring an unbiased evaluation of model predictive performance.
- A **random_state** value of 42 was used to ensure reproducibility of results, meaning the same data split can be obtained consistently across multiple runs.

## 5.5 Data Scaling

- Feature scaling was performed using StandardScaler to standardize the range of independent variables before model training.
- The scaler was fitted on the training data and then applied to both training and testing datasets to ensure consistent transformation.
- Standardization transforms the data such that each feature has a mean of 0 and a standard deviation of 1, ensuring that all variables are on a comparable scale.
- Scaling was applied because machine learning algorithms such as Logistic Regression, KNN, and SVM are sensitive to differences in feature magnitude, and variables with larger values may dominate the model.
- The test data was only transformed (not fitted) to prevent data leakage, ensuring that information from the test set does not influence the training process.
- The scaled arrays were converted back into DataFrame format to preserve column names, ensuring easier interpretation and consistency in further modeling steps.

# 6. Model Building

## 6.1 Choice of Evaluation Metric for Optimization

In this classification problem, the main objective is to accurately predict whether a hotel booking will be canceled or not canceled, so that the business can make informed decisions regarding room allocation, pricing strategies, and customer management. Incorrect prediction of booking cancellations may lead to revenue loss, inefficient resource utilization, and poor operational planning. Therefore, selecting an appropriate evaluation metric is essential to ensure that the model performs effectively in identifying cancellation patterns.

Accuracy alone is not considered sufficient for evaluating model performance, as it may give misleading results in cases where the dataset contains an unequal distribution of canceled and non-canceled bookings. In such situations, a model may appear to perform well simply by predicting the majority class more frequently, without actually capturing meaningful patterns in the data.

Hence, the F1-score is chosen as the primary metric for optimization because it provides a balance between precision and recall. Precision measures how many of the predicted cancellations are actually correct, ensuring that the model does not generate excessive false alarms. Recall measures how effectively the model is able to identify actual cancellations, which is particularly important for minimizing revenue loss and improving booking management strategies. By considering both false positives and false negatives, the F1-score provides a more comprehensive evaluation of model performance.

Optimizing the F1-score ensures that the model achieves a suitable balance between identifying true cancellations and avoiding incorrect predictions. This leads to more reliable and practical predictions, ultimately helping the business improve planning efficiency, reduce uncertainty, and make better strategic decisions.

## 6.2 Logistic Regression Model

```
Optimization terminated successfully.
         Current function value: 0.589829
         Iterations 88
                           Logit Regression Results
==============================================================================
Dep. Variable:         booking_status   No. Observations:                28476
Model:                          Logit   Df Residuals:                    28452
Method:                           MLE   Df Model:                           23
Date:                Sun, 22 Mar 2026   Pseudo R-squ.:                 0.07138
Time:                        09:06:35   Log-Likelihood:                -16796.
converged:                       True   LL-Null:                       -18087.
Covariance Type:            nonrobust   LLR p-value:                     0.000
==========================================================================================
                                     coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                             -0.8884   3.79e+04  -2.35e-05      1.000   -7.42e+04    7.42e+04
no_of_adults                       0.0738      0.015      4.986      0.000       0.045       0.103
no_of_children                     0.0097      0.018      0.538      0.590      -0.026       0.045
no_of_weekend_nights               0.0665      0.013      5.062      0.000       0.041       0.092
required_car_parking_space        -0.2596      0.020    -13.089      0.000      -0.298      -0.221
arrival_year                       0.5219      0.017     30.142      0.000       0.488       0.556
arrival_month                      0.1209      0.014      8.777      0.000       0.094       0.148
arrival_date                       0.0069      0.013      0.524      0.600      -0.019       0.033
repeated_guest                    -0.2893      0.062     -4.691      0.000      -0.410      -0.168
no_of_previous_cancellations       0.0963      0.023      4.262      0.000       0.052       0.141
no_of_previous_bookings_not_canceled -0.5926   0.274     -2.166      0.030      -1.129      -0.056
avg_price_per_room                 0.1208      0.022      5.485      0.000       0.078       0.164
type_of_meal_plan_Meal Plan 2      0.2848      0.014     20.621      0.000       0.258       0.312
type_of_meal_plan_Not Selected    -0.1000      0.014     -7.079      0.000      -0.128      -0.072
room_type_reserved_Room_Type 2    -0.0015      0.013     -0.116      0.907      -0.027       0.024
room_type_reserved_Room_Type 3     0.0053      0.013      0.417      0.676      -0.020       0.030
room_type_reserved_Room_Type 4    -0.0884      0.015     -6.041      0.000      -0.117      -0.060
room_type_reserved_Room_Type 5    -0.0070      0.014     -0.514      0.607      -0.034       0.020
room_type_reserved_Room_Type 6     0.0231      0.018      1.314      0.189      -0.011       0.058
room_type_reserved_Room_Type 7    -0.0347      0.013     -2.576      0.010      -0.061      -0.008
market_segment_type_Complementary -1.0907   1.28e+06  -8.54e-07      1.000   -2.5e+06    2.5e+06
market_segment_type_Corporate     -0.1859      0.056     -3.305      0.001      -0.296      -0.076
market_segment_type_Offline       -0.1242      0.104     -1.190      0.234      -0.329       0.080
market_segment_type_Online         0.0661      0.109      0.606      0.545      -0.148       0.280
==========================================================================================
```

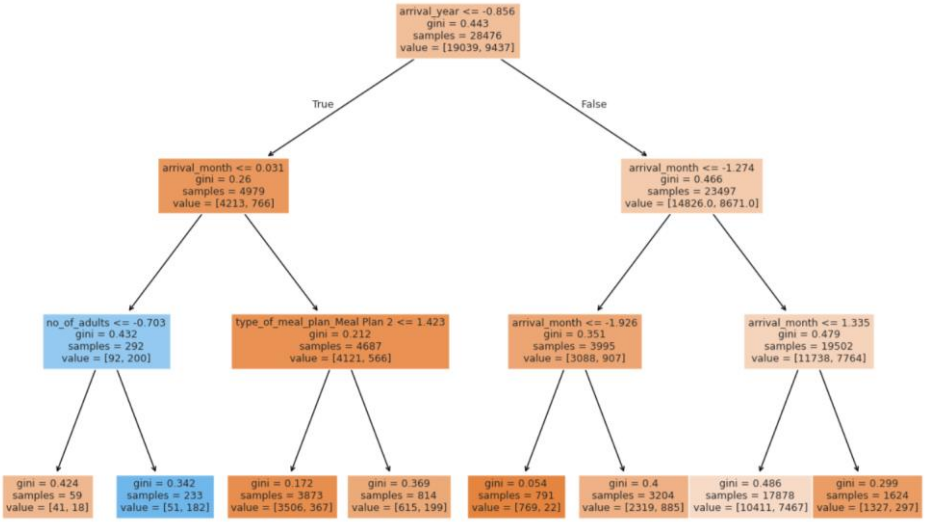
```
Accuracy: 0.6710212108442197
Confusion Matrix:
 [[4538  183]
 [2159  239]]
Classification Report:
              precision    recall  f1-score   support

           0       0.68      0.96      0.79      4721
           1       0.57      0.10      0.17      2398

    accuracy                           0.67      7119
   macro avg       0.62      0.53      0.48      7119
weighted avg       0.64      0.67      0.58      7119
```

- The Logistic Regression model successfully converged after 88 iterations, indicating that the optimization process was completed efficiently and the model parameters were estimated reliably.
- The **overall accuracy** of the model **67.10%**, which suggests that the model is able to correctly classify a moderate proportion of booking outcomes.
- The **F1-score** for class 0 (Not canceled) is **0.79**, which indicates that the model performs well in predicting bookings that are not canceled.
- The **recall value** for class 0 is **0.96**, showing that the model is highly effective in identifying bookings that will not be canceled.
- The **F1-score** for class 1 (Canceled) is **0.17**, which indicates that the model has relatively weak performance in identifying booking cancellations.
- The **recall** for class 1 is only **0.10**, suggesting that the model fails to correctly identify a large proportion of actual cancellations.
- The **confusion matrix** shows that **2159 canceled bookings** were incorrectly predicted as not canceled, indicating a high number of false negatives.
- The model demonstrates class imbalance impact, where predictions are biased toward the majority class (not canceled bookings).
- Several variables such as **lead_time**, **avg_price_per_room**, **arrival_year**, **arrival_month, repeated_guest**, and **no_of_previous_cancellations**, have statistically significant coefficients (p-value < 0.05), indicating that they strongly influence booking cancellation behavior.
- The negative coefficient of **repeated_guest** suggests that repeat customers are less likely to cancel bookings, indicating higher customer reliability.
- The positive coefficient of **lead_time** indicates that bookings made far in advance have a higher probability of cancellation.
- The model shows that pricing-related and historical behavior variables play an important role in predicting cancellations.
- Overall, while the Logistic Regression model provides interpretable results and identifies key influencing factors, its performance in predicting cancellations is limited, suggesting that more advanced classification models may provide better predictive accuracy.

## 6.3 Decision Tree Classifier




```
Accuracy: 0.666385728332631

Confusion Matrix:
 [[4706   15]
 [2360   38]]

Classification Report:
              precision    recall  f1-score   support

           0       0.67      1.00      0.80      4721
           1       0.72      0.02      0.03      2398

    accuracy                           0.67      7119
   macro avg       0.69      0.51      0.41      7119
weighted avg       0.68      0.67      0.54      7119
```

- The Decision tree model achieves an overall accuracy of 66.64%, which is similar to the Logistic regression model, indicating moderate predictive performance.
- The **recall** for class 0 (Not Canceled) is **1.00**, showing that the model is extremely effective in identifying bookings that are not canceled.
- The **F1-score** for class 0 is **0.80**, indicating strong performance in predicting the majority class.
- The **recall** for class 1 (Canceled) is only **0.02**, which means the model is unable to correctly identify most of the actual cancellations.
- The **confusion matrix** shows that **2360** canceled bookings were incorrectly classified as not canceled, indicating a high number of false negatives.
- The **F1-score** for class 1 is **0.03**, which reflects very poor predictive performance for identifying booking cancellations.
- The model appears to be biased towards the majority class, predicting most bookings as not canceled, likely due to class imbalance in the dataset.
- From the decision tree structure, **arrival_year** and **arrival_month** appear as top splitting variables, indicating that time-related factors significantly influence booking cancellation behavior.
- Variables such as **type_of_meal_plan**, **no_of_adults**, and **arrival_month** are also used in decision splits, suggesting that customer characteristics and booking timing affect cancellation probability.
- The low depth of the tree suggests a relatively simple model structure, which may not capture complex relationships between variables.
- Compared to Logistic Regression, the Decision Tree provides visual interpretability, making it easier to understand decision rules influencing booking outcomes.
- However, due to very low recall for canceled bookings, the model is not reliable for identifying customers likely to cancel, which limits its usefulness for proactive business decision-making.
- Overall, the Decision Tree model demonstrates interpretability but does not effectively capture patterns for the minority class, indicating that further tuning or more advanced models may be required for improved performance.

## 6.4 Insights on tuned model performance across different metrics

```
Best Parameters: {'max_depth': 10, 'min_samples_leaf': 1, 'min_samples_split': 10}
Accuracy: 0.6957437842393595
Confusion Matrix:
 [[4511  210]
 [1956  442]]
Classification Report:
              precision    recall  f1-score   support

           0       0.70      0.96      0.81      4721
           1       0.68      0.18      0.29      2398

    accuracy                           0.70      7119
   macro avg       0.69      0.57      0.55      7119
weighted avg       0.69      0.70      0.63      7119

ROC-AUC Score: 0.7426894437732212
Precision: 0.6779141104294478
Recall: 0.18432026688907424
F1 Score: 0.2898360655737705
```

The tunes Decision Tree model shows an improvement in predictive performance after optimizing hyperparameters such as max_depth = 10, min_samples_split = 10, and min_samples_leaf = 1, indicating that the model structure is now better suited to capture patterns in the dataset. The overall accuracy of 69.57% suggests that the model correctly predicts a majority of booking outcomes, showing a slight improvement compared to the base models.

The precision for class 1 (Canceled) is 0.68, indicating that when the model predicts a booking will be canceled, it is correct around 68% of the time. This reflects a reasonable ability of the model to avoid excessive false alarms. However, the recall for class 1 is 0.18, which indicates that the model is still unable to identify a large proportion of actual cancellations, meaning many canceled bookings are incorrectly predicted as not canceled. The F1-score of 0.29 for canceled bookings shows that although the tuned model performs better than the untuned Decision Tree, there is still room for improvement in balancing precision and recall for the minority class. For class 0 (Not Canceled), the F1-score is 0.81, showing strong predictive capability for identifying bookings that will not be canceled. The ROC-AUC score of 0.74 indicates that the tuned model has a good ability to distinguish between canceled and non-canceled

bookings across different probability thresholds. This suggests that the model captures meaningful patterns in the data, even though recall for cancellations remains relatively low.

The macro and weighted averages indicate a moderate balance between classes, showing that hyperparameter tuning has improved the model's generalization ability. Overall, the tuned model demonstrates better predictive stability and improved discrimination capability compared to the base Decision Tree model, making it more reliable for supporting business decisions related to cancellation prediction, although further improvement in recall for canceled bookings may enhance its effectiveness.

# 7. Model Performance & Final Model Selection

## 7.1 Model Performance

| | Model | Accuracy | Precision | Recall | F1 Score | ROC-AUC |
|---|---|---|---|---|---|---|
| 0 | Logistic Regression | 0.671 | 0.566 | 0.100 | 0.170 | 0.655 |
| 1 | Decision Tree | 0.756 | 0.663 | 0.562 | 0.608 | 0.780 |
| 2 | Tuned Decision Tree | 0.696 | 0.678 | 0.184 | 0.290 | 0.743 |

- The Logistic Regression model achieved an accuracy of 67.1%, but shows very low recall (10%) and F1-score (0.17) for canceled bookings, indicating that the model struggles to correctly identify customers who are likely to cancel.
- The Decision Tree model shows the highest accuracy of 75.6% among all models, along with a strong recall of 56.2%, suggesting that it is comparatively more effective in identifying actual booking cancellations.
- The F1-score of 0.608 for the Decision Tree model indicates a good balance between precision and recall, reflecting better overall classification performance compared to the other models.
- The Tuned Decision Tree model achieved an accuracy of 69.6%, with improved precision (0.678) compared to Logistic Regression, indicating better reliability when predicting cancellations.
- However, the recall of the Tuned Decision Tree model (18.4%) is lower than the base Decision Tree model, suggesting that tuning improved precision but reduced the model's ability to detect all actual cancellations.
- The ROC-AUC score is highest for the Decision Tree (0.780), followed by the Tuned Decision Tree (0.743) and Logistic Regression (0.655), indicating that the Decision Tree model has comparatively better capability to distinguish between canceled and non-canceled bookings.
- Overall, model performance varies across metrics, highlighting the trade-off between precision and recall, and indicating that different models capture booking behavior patterns with varying effectiveness.

## 7.2 Final Model Selection

Based on the comparative analysis of multiple classification models, the Decision tree model is selected as the final model for predicting hotel booking cancellations. The selection is based on the model's overall performance across key evaluation metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC, which collectively indicate the effectiveness of the model in identifying both canceled and non-canceled.

Among the evaluated models, the Decision Tree achieved the highest accuracy (75.6%), demonstrating its strong ability to correctly classify booking outcomes. More importantly, the model achieved a significantly higher recall value (56.2%) for canceled bookings compared to Logistic Regression and the Tuned Decision Tree model. Since recall measures the model's ability to correctly identify actual cancellations, this metric is particularly important from a business perspective, as failure to detect cancellations may lead to revenue loss, inefficient room allocation, and poor operational planning.

The Decision Tree model also achieved the highest F1-score (0.608), indicating a good balance between precision and recall. Additionally, the model recorded the highest ROC-AUC score (0.780), suggesting that it has a strong capability to distinguish between canceled and non-canceled bookings across different probability thresholds. These results indicate that the Decision Tree model is able to capture meaningful patterns in customer booking behavior more effectively than the other models. Another advantage of the Decision tree model is its interpretability, as the decision rules can be easily visualized and understood. This aids in identifying key factors influencing booking cancellations, such as arrival month, pricing, previous cancellations, and customer characteristics, enabling better strategic decision-making.

Therefore, considering both predictive performance and interpretability, the **Decision Tree model** is selected as the final model, as it provides a reliable and practical approach for predicting booking cancellations and supporting business decisions aimed at reducing cancellation risk and improving operational efficiency.

# 8. Actionable Insights

The analysis of hotel booking data provides several actionable insights that can directly support business decision-making and help reduce booking cancellations. Customers with longer lead times show a higher tendency to cancel bookings, indicating that early reservations are associated with greater uncertainty in travel plans. An important insight from the analysis is the strong influence of pricing and stay patterns on customer decision-making. The variation observed in the average price per room indicates that customers are sensitive to price fluctuations, and higher prices may increase the likelihood of booking cancellations, especially when customers find better alternatives or experience changes in travel plans. This suggests that the business should adopt dynamic pricing strategies that adjust room rates based on seasonality, demand levels, booking time, and customer segments. Customers booking longer weekday stays or weekend stays also show different booking behaviors, which may be associated with business travel, vacations, or special occasions. Understanding these stay patterns can help the business design targeted packages such as discounted long stay offers, weekday deals, or corporate booking benefits to encourage confirmed bookings and reduce cancellations. Additionally, monitoring price trends in relation to cancellation patterns can help the business maintain competitive pricing while ensuring profitability.

Additionally, insights obtained from customer booking patterns highlight the importance of understanding customer preferences and behavioral trends to reduce cancellation risk. The analysis indicates that repeat guests are generally more reliable and less likely to cancel bookings, suggesting that strengthening customer relationships can significantly improve booking stability. The business can focus on developing loyalty programs, reward points, exclusive member benefits, and personalized offers to encourage customers to choose the same hotel for future stays. Providing a consistent and satisfactory customer experience, including efficient service, easy booking modifications, and responsive customer support, can further increase trust and long-term engagement. Moreover, customers who make special requests often demonstrate higher involvement in the booking process, which indicates the need for better service customization and attention to individual preferences. By maintaining detailed customer profiles and offering tailored services such as preferred room types, flexible check-in options, or customized amenities, the business can enhance customer satisfaction and reduce the likelihood of cancellations. Understanding these behavioral patterns allows the company to shift its focus reactive cancellation management to proactive relationship-building strategies, ultimately helping in improving customer loyalty, strengthening brand reputation, and achieving more stable and predictable booking outcomes.

The final Decision Tree model can be effectively used as a decision-support tool to identify bookings that have a higher probability of cancellation well in advance. By predicting cancellation risk early, hotel management can take proactive measures such as optimizing room allocation, adjusting pricing strategies, and implementing controlled overbooking policies to minimize revenue loss. The model can also help the business design targeted promotional offers, discounts, or personalized communication strategies for customers who are more likely to cancel, thereby increasing the chances of booking confirmation. Additionally, these predictive insights allow better planning of operational resources such as staff allocation, room availability, and service preparation, leading to improved efficiency in hotel operations. By integrating the model into the booking management system, the business can continuously monitor customer booking patterns, respond strategically to potential cancellations, and enhance overall customer experience, ultimately contributing to improved profitability and more informed decision-making.

# 9. Business Recommendations

Based on the comprehensive analysis of booking patterns and predictive modeling results, several strategic recommendations can be implemented to reduce booking cancellations and enhance overall business performance. One of the key findings indicates that customers with longer lead times tend to have a higher probability of cancelling their bookings, as plans made far in advance are more likely to change due to personal, professional, or travel-related uncertainties. To address this, the business should introduce structured engagement strategies such as automated booking reminders, periodic follow-up emails, and timely confirmation messages that reinforce customer commitment. Additionally, implementing flexible rescheduling options or providing incentives for maintaining confirmed bookings can encourage customers to retain their reservations. Introducing partial advance payment requirements or tiered cancellation policies for bookings made well in advance can further discourage casual reservations and reduce last-minute cancellations, thereby improving revenue predictability.

Pricing strategy also plays a significant role in influencing customer booking decisions. The observed variation in average room price suggests that customers are sensitive to price changes and may cancel bookings if they find better alternatives or perceive lower value for the price paid. Therefore, the business should adopt dynamic pricing strategies that adjust room rates based on demand patterns, seasonal trends, and competitor pricing, and customer segments. Offering early-bird discounts for advance bookings, special promotional packages during low-demand periods, and bundled service offers can make bookings more attractive and increase customer commitment. Additionally, offering discounted rates for longer stays, corporate bookings, or off-season reservations can help maintain occupancy levels while improving customer satisfaction. A balanced pricing strategy that aligns perceived value with service quality can significantly improve booking stability and reduce cancellation probability.

Customer behavior analysis also highlights the importance of strengthening relationships with repeat customers. Repeat guests tend to demonstrate higher trust in the service and show lower cancellation tendencies, indicating that customer loyalty plays a crucial role in ensuring consistent bookings. The business should focus on developing structured loyalty programs that provide benefits such as reward points, exclusive discounts, priority services, and personalized offers for returning customers. Maintaining a centralized customer database can help the organization track preferences, booking frequency, and service expectations, allowing the business to deliver customized experiences. Personalized communication, such as special offers on birthdays or anniversary stays, tailored room preferences, or targeted promotional messages, can enhance customer engagement and build long-term relationships. By improving customer satisfaction and strengthening brand loyalty, the business can create a stable customer base that contributes to predictable revenue generation.

Another important recommendation is related to customers who have a history of previous cancellations, as such behavior often indicates a higher likelihood of future cancellations. For these customers, the business may consider implementing slightly stricter booking conditions, such as mandatory confirmation reminders, limited free cancellation windows, or small advance deposits to ensure commitment. At the same time, customers making special requests often show greater involvement in the booking process, which suggests that fulfilling these requests efficiently can enhance satisfaction and reduce uncertainty regarding the stay. Providing customized services such as flexible check-in/check-out options, preferred room arrangements, or additional amenities can improve the overall customer experience and encourage booking confirmation. Improving service responsiveness and addressing customer concerns promptly can also reduce the probability of cancellations caused by dissatisfaction or lack of clarity.

The implementation of the final Decision Tree predictive model provides an opportunity for the business to adopt a data-driven approach to cancellation management. By identifying high-risk bookings in advance, hotel management can take proactive actions such as adjusting inventory allocation, applying controlled overbooking strategies, and offering targeted incentives to customers who are more likely to cancel. The model can also assist in demand forecasting, helping the business optimize staff allocation, room availability planning, and pricing adjustments based on predicted booking trends. Integrating predictive analytics into the reservation system allows continuous monitoring of booking patterns, enabling timely strategic interventions that reduce uncertainty and improve operational efficiency. Overall, the use of predictive insights supports better planning, improves resource utilization, enhances customer satisfaction, and ultimately contributes to improved profitability and sustainable business growth.

# 10. Conclusion

This project focused on applying classification techniques to predict hotel booking cancellations in order to support more effective and data-driven business decision-making. Through detailed exploratory data analysis, several important factors influencing customer booking behavior were identified, including lead time, average price per room, previous cancellation history, number of successful previous bookings, and number of special requests. The analysis revealed that customer decisions are influenced by a combination of pricing factors, booking timing, and past behavioral patterns. Data preprocessing steps such as outlier treatment using the IQR method, encoding of categorical variables, and feature scaling ensured that the dataset was well-structured and suitable for building reliable machine learning models.

Multiple classification models were developed and evaluated using key performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score. The comparative analysis showed that different models capture booking patterns with varying levels of effectiveness. Among the evaluated models, the Decision Tree classifier demonstrated the most balanced performance, particularly in identifying booking cancellations, which is an important requirement from a business perspective. The model was able to capture meaningful relationships between customer characteristics and booking outcomes, making it a suitable tool for predicting cancellation risk. Hyperparameter tuning further improved model stability and helped ensure better generalization of the predictive patterns.

The results of this study highlight the importance of using predictive analytics to understand customer behavior and reduce uncertainty in booking management. By identifying bookings that are more likely to be canceled, the business can take proactive steps such as adjusting pricing strategies, implementing customer retention initiatives, and optimizing room allocation. These insights also help in improving operational planning, resource utilization, and customer service strategies. Integrating the predictive model into the booking system can support continuous monitoring of booking trends and enable timely strategic interventions. Overall, the use of machine learning techniques in this project demonstrates how data-driven insights can enhance decision-making, improve efficiency, reduce potential revenue loss, and contribute to long-term business growth and sustainability.